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Systematic review: Comparing zk-SNARK, zk-STARK, and bulletproof protocols for privacy-preserving authentication

Bjorn Oude Roelink | Mohammed El-Hajj  | Dipti SarmahSCS, EEMCS, University of Twente,
Enschede, Netherlands**Correspondence**Mohammed El-Hajj, SCS, EEMCS,
University of Twente, Enschede,
Netherlands.
Email: m.elhajj@utwente.nl**Abstract**

This systematic literature review examines the implementation and analysis of zk-SNARK, zk-STARK, and bulletproof non-interactive zero-knowledge proof (NIZKP) protocols in privacy-preserving applications across diverse sectors. Examining 41 research works obtained through the systematic search queries and filtering criteria published from 2015 to April 2023, we categorized findings into financial, medical, business, general, and other domains. Our analysis highlights significant variations of up to several orders of magnitude in real-world performance across implementations utilizing NIZKP protocols. However, divergent methodologies in security analyses hindered conclusive comparisons. Addressing research gaps, our future endeavors aim to establish a real-world benchmark for these protocols.

KEYWORDS

bulletproof, performance, privacy-preserving, security analysis, zero knowledge, zk-SNARK, zk-STARK

1 | INTRODUCTION

Zero-knowledge proofs (ZKPs) represent a relatively recent cryptographic innovation, initially introduced by Goldwasser et al.¹ They enable a prover to demonstrate certain knowledge to a validator without disclosing the specifics of that knowledge. However, traditional ZKPs are interactive, necessitating multiple exchanges between the prover and verifier for trust or rejection. Additionally, these proofs cannot be reverified by other parties without initiating fresh interactions, limiting their practicality. Addressing this limitation, Blum et al. proposed Non-Interactive Zero-Knowledge Proofs (NIZKPs).² NIZKPs empower a verifier to confirm a claim in a single interaction, allowing subsequent verifiers to validate the same claim later.

Notably, ZKPs, especially the non-interactive variants, have gained prominence in cryptocurrencies like ZCash and Ethereum.³ In these contexts, they facilitate transaction verification without disclosing sensitive transaction details, preserving privacy. Although cryptocurrencies have significantly bolstered the visibility of ZKPs due to the surge in blockchain technologies, their utility extends beyond this domain.

Traditional authentication methods, including password-based systems and token-based authentication, have long been the cornerstone of digital security. However, these methods are plagued by significant limitations and vulnerabilities in the face of evolving cyber threats. Password-based authentication,⁴ despite its widespread use, is

All authors are equally contributing for this article.

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inherently vulnerable. Users tend to choose weak passwords or reuse them across multiple accounts, making it easy for attackers to gain unauthorized access through techniques like brute-force attacks. Moreover, passwords can be easily intercepted or stolen through phishing attacks, keyloggers, or data breaches, leading to compromised accounts and sensitive data leaks. Token-based authentication,⁵ while more secure than simple passwords, is not immune to challenges. Physical tokens can be lost or stolen, compromising the authentication process. Additionally, time-based tokens are susceptible to interception during transmission, allowing attackers to hijack the authentication session.

Cyber-attacks have become increasingly sophisticated, leveraging advanced techniques such as machine learning algorithms and artificial intelligence to bypass traditional security measures. Attackers employ tactics like social engineering, malware, and phishing schemes that specifically target authentication processes. As a result, organizations face a constant struggle to stay one step ahead of malicious actors, requiring innovative and adaptive security solutions. In light of these challenges, there is a pressing need for authentication mechanisms that are not only highly secure but also respect user privacy. Privacy-preserving authentication⁶ methods ensure that sensitive user information is not unnecessarily disclosed during the authentication process, safeguarding individuals' privacy rights. Such mechanisms often employ advanced cryptographic techniques, including zero-knowledge proofs, enabling users to prove their identities without revealing sensitive data. Authentication represents a sphere where zero-knowledge proofs can revolutionize practices. ZKPs empower users to prove claims without revealing the intricacies of the proof, ensuring privacy during authentication.⁶ Furthermore, ZKPs find applications in age verification, enabling users to prove they meet a specific age requirement without disclosing their exact birth date,⁷ as well as in the "know your customer" process, where banks need income information without requiring the exact figure.

As the digital landscape continues to evolve, addressing these challenges becomes increasingly more important. The development and adoption of innovative, secure, and privacy-preserving authentication mechanisms are crucial to mitigating the risks posed by sophisticated cyber threats and ensuring a safer digital environment for individuals and organizations alike. Privacy-preserving authentication is an important area of research in cryptography,⁸ with applications in anonymous credentials, secure voting protocols, and other areas where privacy is a concern. ZKPs are a powerful tool for privacy-preserving authentication, allowing one party to prove to another party that they know a certain piece of information without revealing the information itself. In recent years, several types of zero-knowledge proofs have emerged as leading contenders for practical use: zk-SNARKs,⁹ zk-STARKs,¹⁰ and bulletproofs.¹¹ These protocols differ in their efficiency, security, and underlying mathematical techniques, making them suitable for different types of applications. In the rapidly evolving landscape of zero-knowledge proofs and privacy-preserving authentication, there exist notable research gaps and challenges that necessitate a comprehensive review. Existing literature, while substantial, often lacks a unified synthesis and critical analysis.

One significant research gap lies in the exploration of practical implementations of zero-knowledge proofs across diverse applications that use zk-SNARKs, zk-STARKs, and bulletproof. While theoretical frameworks abound, there is a scarcity of empirical studies detailing real-world applications and their effectiveness. Additionally, the intersection of zero-knowledge proofs and emerging technologies, such as blockchain and IoT, presents unexplored avenues for research. Understanding the integration challenges and optimizing these technologies for real-time applications is an underexplored area. In the realm of privacy-preserving authentication, the integration of usability and security remains a challenge. Striking a balance between robust security measures and user-friendly experiences is a persistent gap. Furthermore, there is a scarcity of research addressing the vulnerabilities of existing privacy-preserving authentication methods to novel cyber threats. As attackers continuously adapt their tactics, it is crucial to identify potential weaknesses and develop countermeasures.

This systematic review of the literature is motivated by the unique properties and functionalities of zero-knowledge proofs (ZKPs) in the context of privacy-preserving authentication when compared to other widely used technologies such as Homomorphic Encryption, Pseudonymization, and Secure Multi-Party Computation (MPC). ZKPs offer a novel approach to privacy preservation, fundamentally distinct from conventional methods. Unlike traditional techniques that often involve data transformation or distribution among parties, ZKPs enable verification of the validity of a statement without revealing any underlying information. This unique characteristic not only ensures privacy but also minimizes computational overhead and communication complexities inherent in other methodologies. Moreover, ZKPs introduce a shift in the perspective of system design and implementation by eliminating the need for trust between parties. Unlike some encryption-based techniques where decryption keys or partial information must be shared, ZKPs allow parties to

interact securely without exchanging sensitive data, thereby reducing the risk of data breaches and unauthorized access. In contrast to typical privacy preservation technologies, ZKPs offer unparalleled advantages in scenarios where data confidentiality, integrity, and authenticity are of vital importance. Traditional methods often rely on cumbersome encryption or data obfuscation techniques, which may not be suitable for all use cases, especially those requiring high levels of transparency and auditability. The inherent properties of ZKPs, such as provability, efficiency, and scalability, make them indispensable for modern privacy-preserving authentication systems. By leveraging ZKPs, organizations can achieve robust security guarantees while maintaining user privacy and data confidentiality effectively. Thus, this systematic literature review aims to delve into the specific applications and methodologies of ZKPs, including zk-SNARKs, zk-STARKs, and bulletproofs, within the context of privacy-preserving authentication. Through a comprehensive analysis of existing literature, we seek to outline the unique contributions of ZKPs in addressing contemporary challenges and advancing the landscape of privacy-enhancing technologies. Ultimately, this review aims to provide actionable insights for researchers and practitioners, facilitating informed decision-making and fostering innovation in the field of privacy-preserving authentication.

The structure of this article is as follows: Section 2 will design the systematic literature review (SLR) to discover the recent academic landscape regarding the implementation of zk-SNARKs, zk-STARKs, and bulletproofs in the context of privacy-preserving authentication. In Section 3, we will conduct all steps of this SLR, and report their results. Next, Section 4 will summarize all articles that are the result of the SLR in the previous section. Section 5 will discuss the results achieved from the SLR and the benchmark done, and give a number of recommendations on the implementation of zk-SNARKs, zk-STARKs, and bulletproofs in the context of privacy-preserving authentication. Finally, Section 5.4 concludes the work done.

2 | DESIGNING THE SYSTEMATIC LITERATURE REVIEW

A SLR entails a thorough analysis of existing research within a clearly defined domain, employing systematic techniques to identify, select, and evaluate relevant articles while gathering and scrutinizing data from these studies. To ensure the rigor of an SLR, it is essential to adopt methods that are both reproducible and transparent.¹² While various types of literature reviews, like exploratory reviews, aim to uncover published theories, empirical data, and research methodologies in academic literature,¹³ our objective extends beyond a surface-level examination. We aim to pinpoint gaps in the current state of the art. Hence, we have opted for an SLR approach in this study. This section outlines the framework used for conducting the systematic literature review. Initially, we will craft a precise search string, employing it to query predetermined online databases. Subsequently, we will sift through articles, discarding irrelevant ones, and analyze the characteristics of the retained works. Lastly, we will outline the identified knowledge gaps and propose potential avenues for future research.

2.1 | Methodology

Since we have chosen to undertake a systematic literature review, it is crucial to present a detailed account of our review process to ensure the reproducibility of our results. Our process begins by integrating the research topic, context, aims, and objectives, leading to the formulation of research questions, as detailed in Section 2.2. In Section 2.3, we outline our literature collection procedure based on these research questions. Here, we craft precise search queries to locate literature relevant to our inquiries and specify the research databases subjected to these queries. Subsequently, in Section 2.4, we clarify the subsequent step involving the filtration of acquired literature. This process includes the removal of duplicate sources, exclusion of materials with no use in our topic and context, and a manual review process. The manual review involves an initial evaluation of abstracts for eligibility and a subsequent scrutiny of full texts. Only works meeting all these criteria are incorporated. Following this, information retrieval techniques are applied to the selected research. The outcomes are evaluated, and a bibliometric analysis is conducted. The final stage encompasses the creation of a comprehensive report. This document enumerates the research findings, addresses each research question, and delves into the obtained results. For a visual representation of the entire systematic literature review, the process is illustrated in Figure 1.

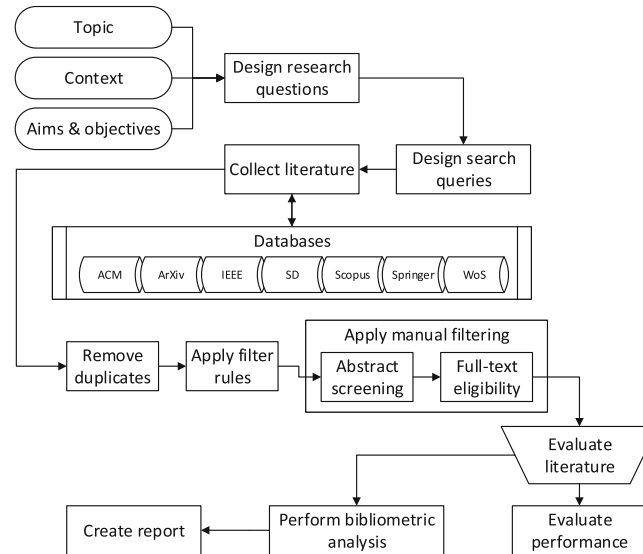


FIGURE 1 Systematic literature review process diagram.

2.2 | Research questions

In this subsection, we outline the key research questions that will be addressed as a result of the SLR. These questions serve as the central points for our investigation into the implementation of zk-SNARKs, zk-STARKs, and bulletproofs in the context of privacy-preserving authentication:

1. What are the existing real-world use case implementations of zk-SNARKs, zk-STARKs, and Bulletproofs in privacy-preserving authentication, as documented in the literature? How are these protocols currently being applied in practical scenarios, and what insights can be drawn from these implementations regarding their effectiveness, challenges faced during deployment, and potential improvements for broader adoption?
2. What are the comparative performance and security implications of implementing zk-SNARKs, zk-STARKs, and bulletproofs in privacy-preserving authentication systems? This includes an analysis of authentication speed, computational overhead, and resilience against various types of attacks, as well as potential vulnerabilities.

These research questions guide our systematic exploration of the literature, aiming to provide comprehensive insights into the practical applications, performance, security, user experience, and challenges associated with zk-SNARKs, zk-STARKs, and bulletproofs in the realm of privacy-preserving authentication.

2.3 | Collecting literature

The SLR scope consists of two dimensions, namely, publication period and publication venues. Concerning the publication period, this SLR includes papers published from 2013 to 2023. The year 2013 was taken as the starting point since, from this year, research in the usage of ZKPs for privacy-preserving authentication started growing. We conducted the literature search in June 2023, accepting only studies until that point. From a publication venue perspective, we searched for studies in the following databases; ScienceDirect, ACM Digital Library, IEEE Xplore, Scopus, ArXiv, Web of Science, and Springer Link. The list of included literature databases and URLs can be found in Table 1.

2.3.1 | Searching query

We seek a comparative analysis among zero-knowledge proof protocols, specifically focusing on zk-STARK, zk-SNARK, and bulletproof protocols. To achieve this objective, our search query encompasses terms ensuring that each retrieved

TABLE 1 Included research databases.

Database	URL
ACM Digital Library	https://dl.acm.org
ArXiv	https://arxiv.org
IEEE Explore	https://ieeexplore.ieee.org
ScienceDirect	https://www.sciencedirect.com
Scopus	https://www.scopus.com
Springer Link	https://link.springer.com
Web of Science	https://www.webofscience.com

work discusses at least one of these protocols. We construct our query using three distinct components, combined using the OR operator:

- ("zk-STARK" OR "zk-STARKs" OR "zero-knowledge Scalable Transparent Argument of Knowledge")
- ("zk-SNARK" OR "zk-SNARKs" OR "zero-knowledge Succinct Non-interactive Argument of Knowledge")
- ("Bulletproof" OR "Bulletproofs")

Additionally, our interest extends to the application of these protocols in the realm of privacy-preserving authentication. To narrow down our search, we require that the following query components align:

- ("privacy-preserving" OR "preserve privacy")
- ("authentication" OR "identity")

Combining these elements, our complete query is as follows:

```
(
("zk-STARK" OR "zk-STARKs" OR
"zero-knowledge Scalable Transparent Argument of Knowledge")
OR ("zk-SNARK" OR "zk-SNARKs" OR
"zero-knowledge Succinct Non-interactive Argument of Knowledge")
OR ("Bulletproof" OR "Bulletproofs")
)
AND ("privacy-preserving" OR "preserve privacy")
AND ("authentication" OR "identity")
```

2.4 | Filtering literature

As mentioned in Section 2.3, we estimate that the collection of literature will comprise hundreds of research publications. Since these works are not yet detailed enough to be evaluated right away, we must first filter them to exclude any publications that have nothing to do with our topic or context. There are two stages to this filtering. Initially, a rule-based filter is applied. Section 2.4.1 has a description of this procedure. We then apply the second stage of manual filtering on the resulting collection, which is comprised of two stages: full-text eligibility assessment, which is described in Section 2.4.3, and title and abstract screening, which is described in Section 2.4.2. Figure 2 shows a PRISMA¹⁴ flow diagram that represents the whole filtering procedure we used for the literature review.

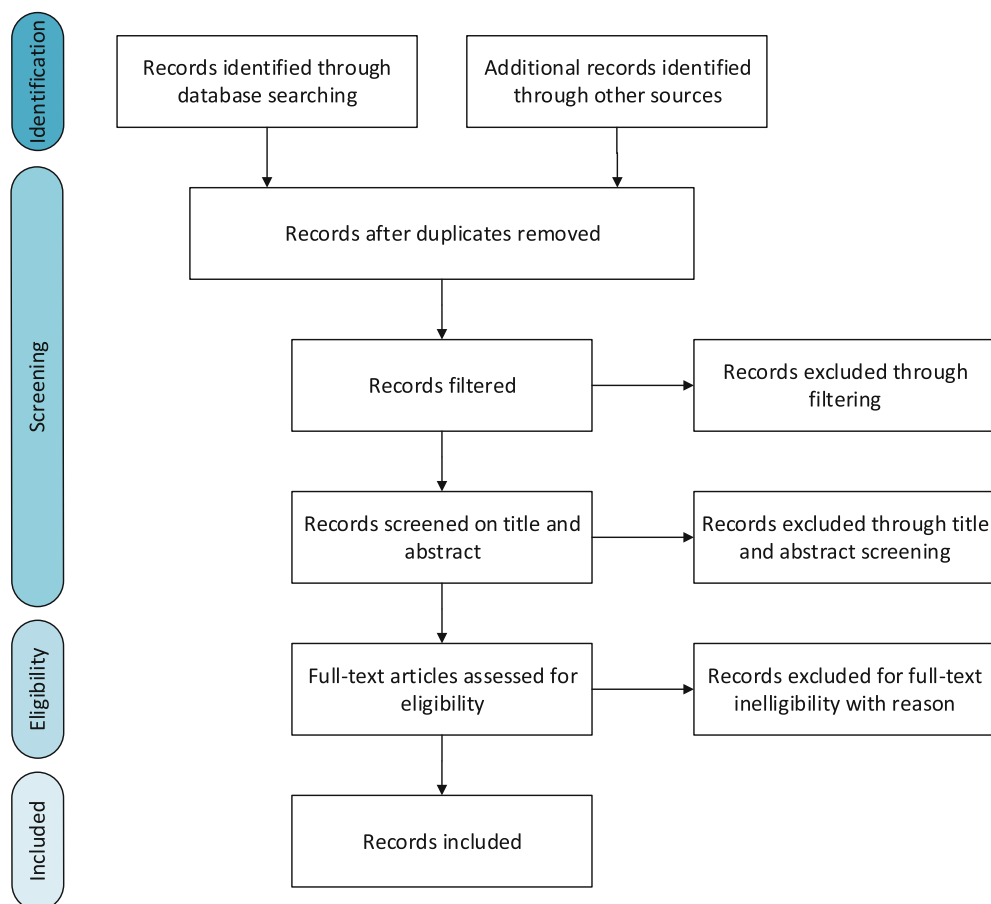


FIGURE 2 Systematic literature review PRISMA flow diagram.

TABLE 2 Literature filter rules.

Filter	Rule
Language	Exclude research written in languages other than English
Content type	Exclude research published in a book or included as a chapter thereof
Publication date	Exclude research published before 2013
Duplicate	Exclude duplicates of a unique research

2.4.1 | Rule-based filtering

Our initial filtering process is conducted through a rule-based filter, which functions based on article metadata, including factors such as content type and publication date. The criteria we have applied to this filter are documented in Table 2. Any article matching one of the exclusion criteria is removed from the collection, while only articles that do not meet any of the exclusion criteria in this filter are retained.

By implementing these rules, we ensured a refined collection that aligns precisely with our research criteria, enhancing the quality and relevance of the literature under consideration.

2.4.2 | Manual title and abstract screening

Following the application of the rule-based filter detailed in Section 2.4.1, we proceed with a manual filtering process involving the examination of each article's title and abstract. Articles that meet our predefined inclusion criteria and do not violate any exclusion criteria are retained, while others are excluded.

Our inclusion criteria encompassed the following aspects:

1. The article must present or describe a use-case where zero-knowledge proofs find application.
2. Privacy preservation should be the primary objective of the article. This focus can extend beyond authentication, covering any privacy-preserving application.
3. The title or abstract of the article must indicate the usage or potential usage of zero-knowledge proofs. Articles that imply the use of zero-knowledge proofs, even if not explicitly mentioned, are included for further assessment during the full-text eligibility evaluation.

Conversely, we employed the following exclusion criteria:

1. Articles categorized as surveys, literature reviews, or overviews are excluded, as our systematic literature review focuses on original research for evaluation rather than third-party evaluations.
2. The main objective of the article should be a privacy-preserving application. Articles lacking this focus are filtered out.
3. Articles introducing new zero-knowledge proof schemes are excluded. Our review specifically examines the usage of three established protocols, namely zk-STARK, zk-SNARK, and bulletproof, in privacy-preserving applications. Works describing new protocols are irrelevant for our SLR as they do not describe any privacy-preserving application using one of the three protocols of interest.
4. Articles proposing concepts (e.g., mathematical ideas) for application in other privacy-preserving techniques are not relevant to our review, as they do not directly demonstrate the application of zero-knowledge proofs in privacy preservation.
5. Articles suggesting alternatives to the use of zero-knowledge proofs for specific applications are also excluded. Our review focuses on comparing the usage of zk-SNARK, zk-STARK, and bulletproof zero-knowledge proof protocols.

These criteria guided our comprehensive evaluation, ensuring the relevance and applicability of the selected articles to our study.

2.4.3 | Manual full-text eligibility assessment

The last filtering step involves a thorough full-text eligibility assessment of the articles shortlisted after the title and abstract screening outlined in Section 2.4.2. We evaluated each article by examining its entire content to determine its eligibility for inclusion in our analysis, adhering to our predetermined inclusion and exclusion criteria.

Similar to the manual title and abstract screening, articles must meet all eligibility criteria to be retained. For those excluded during this assessment, we provide explicit reasons for their exclusion. We established the following eligibility criteria for articles to be included in our final evaluation:

1. **Direct usage of zero-knowledge proof protocols:** The article must directly employ one of the three specified zero-knowledge proof protocols (zk-STARK, zk-SNARK, or bulletproof). Indirect usage, such as through a blockchain that implements zero-knowledge proofs, is insufficient.
2. **In-depth application of zero-knowledge proof protocols:** The usage of zero-knowledge proof protocols should be substantial and profound. Mere surface-level references, like solely suggesting the possibility of using zk-SNARKs, do not meet the criteria.
3. **Primary focus on privacy preservation:** The primary objective of the article should be privacy preservation. The privacy preserving aspect should not just be an unintended consequence.

These tight criteria ensure that only articles deeply embedded in the direct application of zk-STARK, zk-SNARK, or bulletproof protocols, with a strong focus on privacy preservation, are included in our final evaluation, enhancing the relevance and integrity of our study.

2.5 | Presenting knowledge gaps and new research directions

In the final phase, we will have read and analyzed all remaining papers. With this information, we hope to identify knowledge gaps in the literature and possible new research direction.

3 | PERFORMING THE SYSTEMATIC LITERATURE REVIEW

In Section 2, we outlined the systematic literature review process. This section delves into the practical execution of the methodology detailed there, presenting the outcomes of each stage. In Section 3.1, we present the results of our literature collection efforts from various research databases, aligning with the process described in Section 2.3. Moving forward, in Section 3.2, we showcase the outcomes of filtering the discovered articles, adhering to the criteria outlined in Section 2.4. Lastly in Section 3.4, we assess the articles that remain post filtering. Upon concluding this section, we possess the necessary data to present the results of our systematic literature review and engage in a comprehensive discussion, as outlined in Section 4.

3.1 | Collecting literature

In this section, we outline our approach to literature collection. Our process initiates by querying research databases, following the search parameters detailed in Section 2.3.1. The outcomes of our search queries for each database are documented in Table 3. The numbers listed in the *Results* column represent the total findings encompassing various forms of references, including conference proceedings. However, for our systematic literature review, we focus solely on related articles and, therefore, exclude collections like conference proceedings. The count presented in the *Articles* column represents the identified records that align with our research scope.

3.2 | Filtering literature

In the previous Section 3.1, we discussed our process of collecting literature. Now, we move forward by detailing the filtering phase, a crucial step in refining our research scope. Notably, not all literature records retrieved are equally applicable to our study. To enhance the quality of our research and narrow down our focus, we apply filtering strategies as outlined in Section 2.4. Our filtering process begins with rule-based filtering, described in Section 2.4.1. By applying predefined filter rules (as depicted in Table 2), we reduced the initial pool of 280 records to 245, excluding duplicates, specific content types, outdated publications, and non-English articles as presented in Table 4. Next, we perform manual filtering, involving title and abstract screening, followed by a comprehensive full-text eligibility assessment, detailed in Sections 2.4.2 and 2.4.3 respectively. The result of this manual filtering process is presented in Table 5. The entire filtering process as conducted, including the number of literature articles included and excluded in each step, is illustrated in Figure 3.

Upon completing the filtering process, we were left with 43 included research results. These selected works exhibit a balanced distribution between conference papers (56%) and journal articles (44%), reflecting the diversity in scholarly publications as illustrated in Figure 4A. Further analysis of our included literature reveals interesting patterns in terms of source libraries. Significant contributions stem from ACM Digital Library (30%), Springer Link (28%), and ScienceDirect (23%), signifying their prominence in zero-knowledge proof research for privacy-preserving applications as illustrated

TABLE 3 Number of articles identified using the search query.

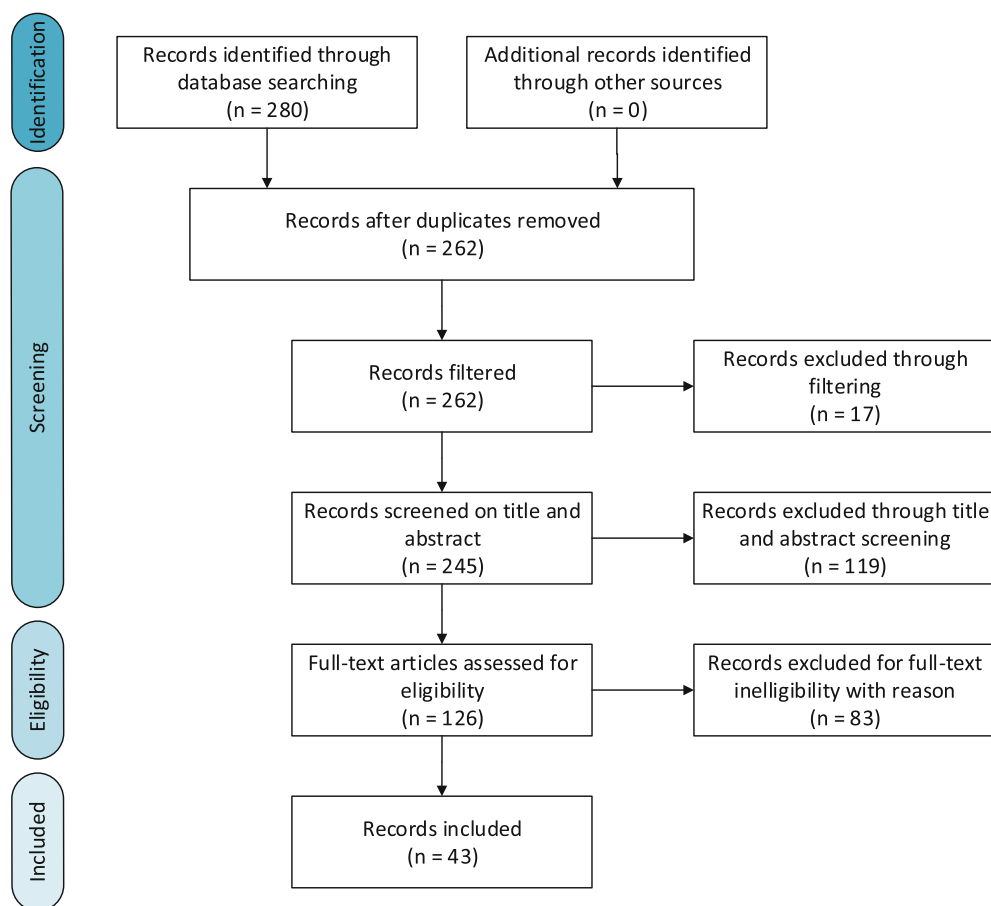
Database	Results	Articles
ACM Digital Library	64	60
ArXiv	0	0
IEEE Explore	5	5
ScienceDirect	76	76
Scopus	19	16
Springer Link	256	114
Web of Science	9	9
Total	429	280

TABLE 4 Number of articles removed through rule-based filtering.

Filter	Filtered	Included	Excluded
Duplicate	280	262	18
Content type	262	246	16
Publication date	246	245	1
Language	245	245	0
Total	280	245	35

TABLE 5 Number of articles removed through manual filtering.

Filter	Filtered	Included	Excluded
Title and abstract screening	245	126	119
Full-text eligibility assessment	126	43	83
Total	245	43	202

**FIGURE 3** Filled systematic literature review PRISMA flow diagram.

in Figure 4B. Notably, the prevalence of certain keywords, such as “blockchain,” underscores the evolving landscape of zero-knowledge proofs, primarily within blockchain technologies. Additionally, the dominance of terms like “privacy,” “zero-knowledge proof,” and “authentication” echoes our research focus as illustrated in Figure 6.

Figure 5 shows the number of included research literature, including the type, by year of publication. We must note that the literature collection was performed on March 31, 2023. This means that the year 2023 in the figure only represents research published in the first quarter of that year. The graph shows a clear increasing trend in the total number of published research on the topic of zero-knowledge proofs for privacy-preserving applications, at least for research that meets our inclusion criteria. This trend can be explained by the young mathematical foundation behind the technology that is ZKPs, next to an increasing interest in its application for privacy-preserving uses in an ever more digital society.

Figure 6 shows the co-occurrence graph of keywords that occur at least twice in the included literature. This graph clearly shows that “blockchain” is the most prevalent keyword among the works included in our systematic literature

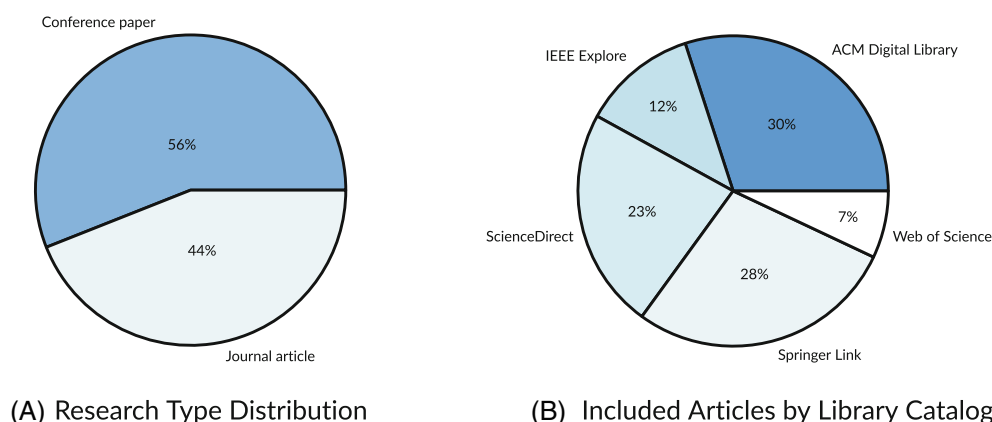


FIGURE 4 Distribution of the included articles. (A) Research type distribution. (B) Included articles by library catalog.

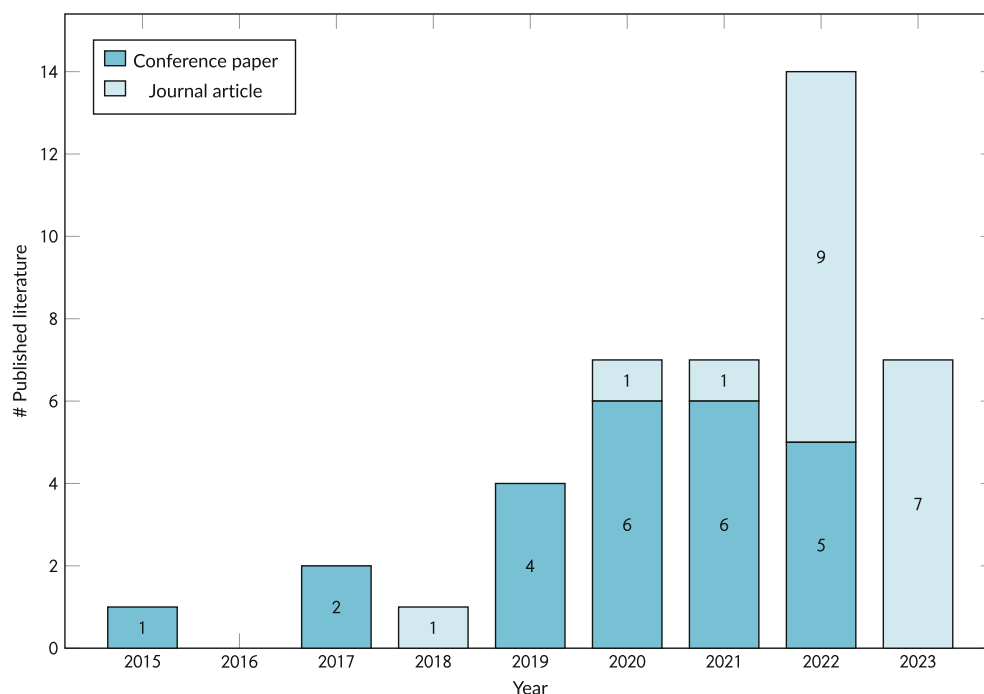


FIGURE 5 Number of research included by year of publication.

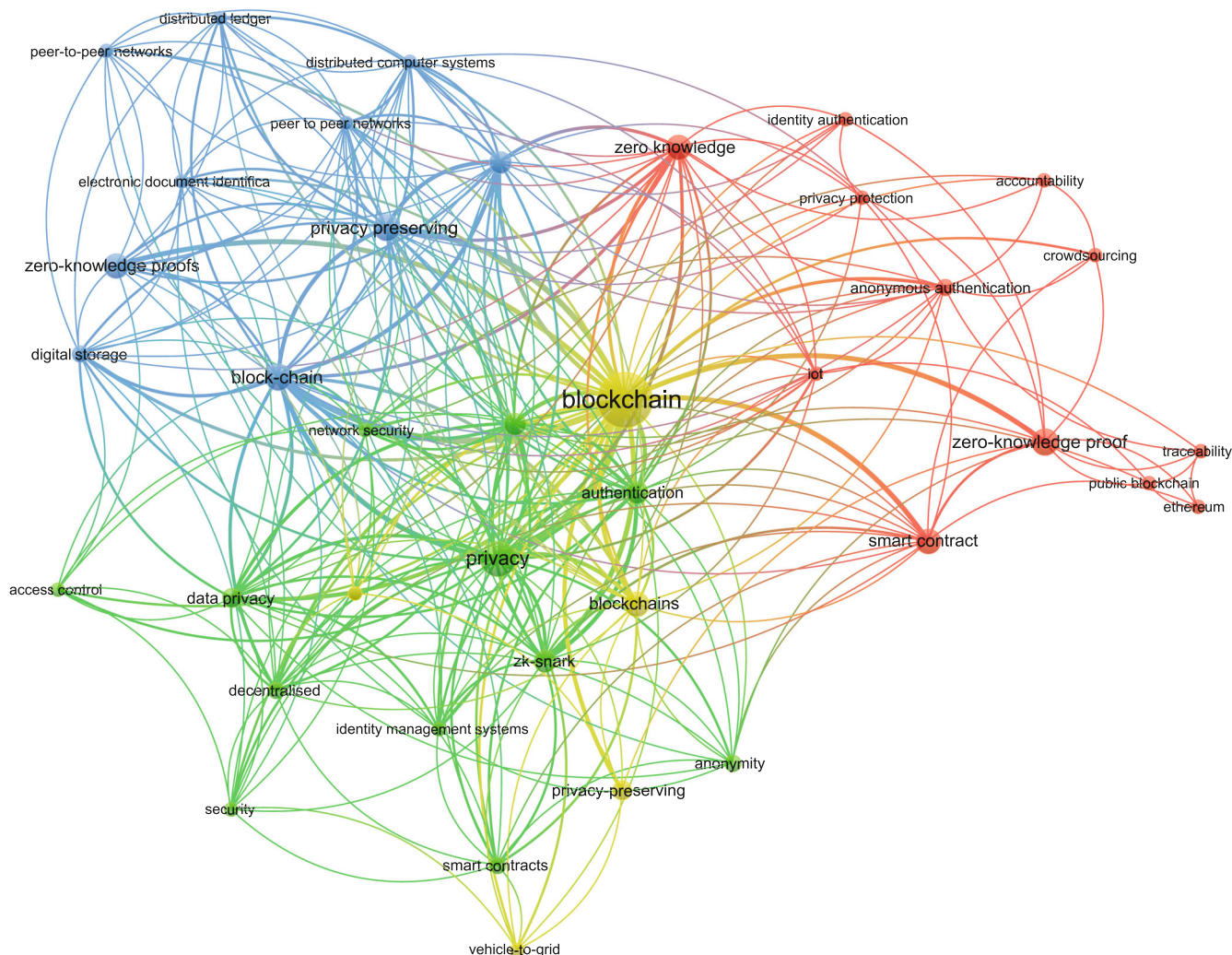


FIGURE 6 Keyword co-occurrence graph of keywords that occur at least twice in the included literature.

review. We can explain this by the main application of zero-knowledge proofs currently being in blockchains, the technology behind cryptocurrencies. The prevalence of this keyword combined with the interest in cryptocurrencies in the past few years also helps explain the rise in the number of research published in the past few years, as seen in Figure 5. The high occurrence of other keywords including “privacy”, “zero-knowledge proof”, and “authentication”, is also logical since these terms are what we searched for when collecting the literature. The three specific zero-knowledge proof protocols “zk-SNARK”, “zk-STARK” and “bulletproofs” are less prevalent, which we can explain by research proposals using just one of these three protocols. Of the three protocols “zk-SNARK” appears the most, which we can explain by it being the oldest and most popular protocol of the three.

Figure 7 shows the co-authorship graph of authors that collaborated with each other, within the group of authors that authored included research works. This graph has 39 clusters, which means that of the 43 included papers and articles, most authors are only connected to authors of the same paper. This seems to indicate that authors of research within our inclusion criteria do not collaborate often. Combined with the results illustrated in Figure 5 however, it can also show that the relevant research field is young and has not had the time to explore follow-up research. Such follow-up research can combine the knowledge of researchers from different domains interested in the topic of zero-knowledge proofs. This limits the degree of connectivity in the co-authorship graph. A third explanation is that the authors of included papers may not be mainly interested in zero-knowledge proofs, but in the different applications which apply zero-knowledge proofs to achieve an objective. Applications being in entirely separate research domains can in that event be a disincentive for collaboration. The latter explanation seems probable given that our search query and filtering criteria look for different applications that apply zero-knowledge proof protocols for privacy preservation.

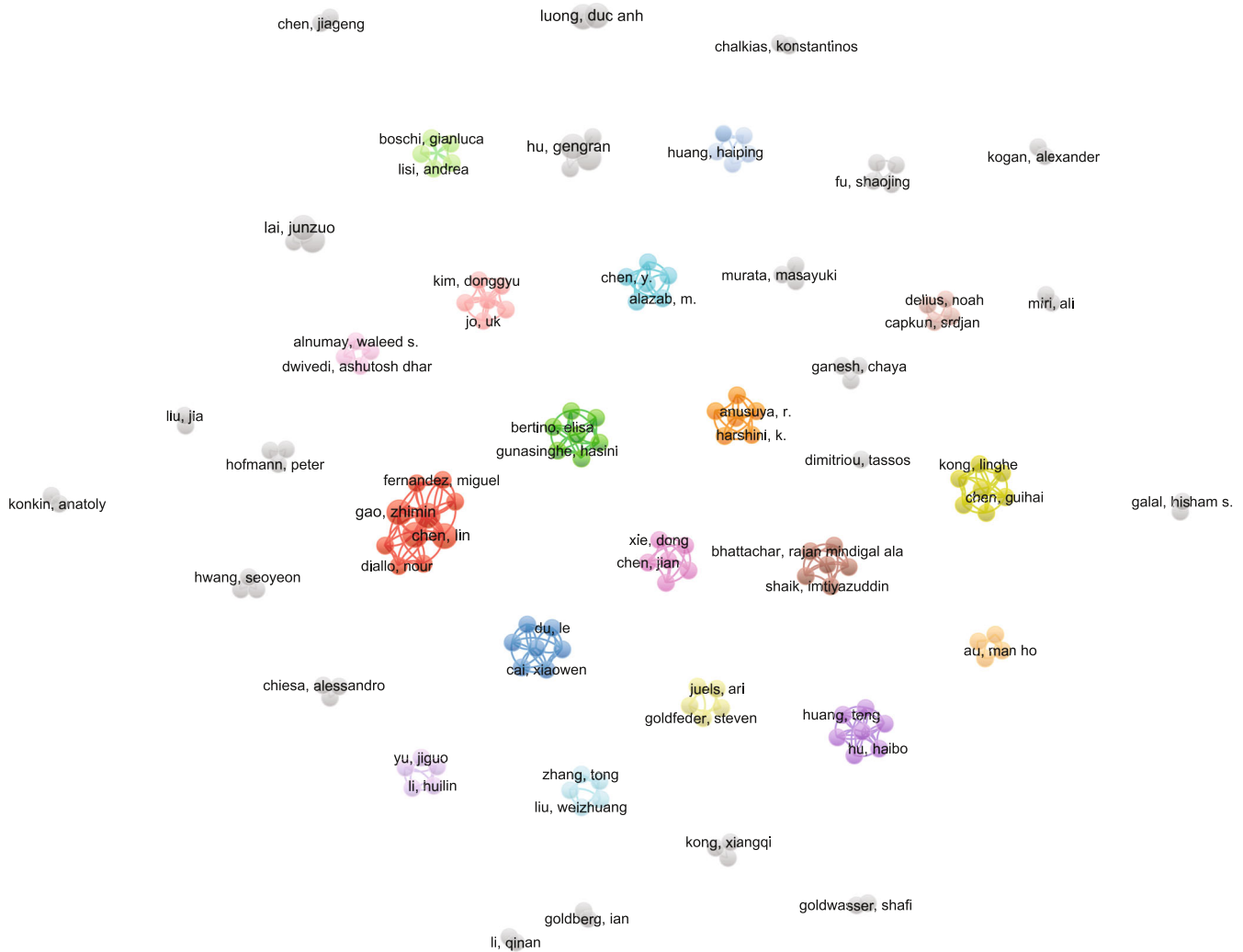


FIGURE 7 Co-authorship graph of included literature.

3.3 | Background

The background section serves as a crucial component of the paper, particularly for non-technical or general readers, as it provides essential context and foundational knowledge necessary to understand the subsequent discussions on zero-knowledge proofs (ZKPs) and their associated protocols, including zk-SNARKs, zk-STARKs, and bulletproofs. By offering a clear and concise overview of ZKPs and their applications, this section aims to clarify and explain complex cryptographic concepts and ensure accessibility for a broader audience. Furthermore, explaining the mathematical notations used in these protocols without overwhelming readers with technical jargon is essential for fostering comprehension and engagement.

1. **zk-SNARKs:** Enables one party to prove to another that they possess certain information without revealing the information itself, all while maintaining zero-knowledge of the information. This property makes zk-SNARKs invaluable for privacy-preserving transactions, such as in blockchain applications. The key cryptographic concepts used in zk-SNARKs are:¹⁵
 - a. **Proof Generation Key and Verification Key:** These keys are generated during the setup phase of the protocol. The proof generation key is used by the prover to generate proofs, while the verification key is used by the verifier to verify the proofs. They play a crucial role in ensuring the integrity and security of the zero-knowledge proofs.

- b. **Zero-Knowledge Proof:** This is the cryptographic proof generated by the prover to convince the verifier of the validity of a statement without revealing any additional information. Zero-knowledge proofs allow for privacy-preserving transactions and ensure that sensitive information remains confidential.
- 2. **zk-STARKs:** Offers zero-knowledge proofs that are both transparent and scalable, making them suitable for verifying computations over large datasets. Unlike zk-SNARKs, zk-STARKs do not require a trusted setup, enhancing their security properties. The key cryptographic concepts used in zk-STARKs are:¹⁶
 - a. **FRI Protocol (Fast Reed-Solomon Interactive Oracle Proof):** The FRI protocol is used in zk-STARKs for polynomial evaluation and interpolation. It enables the prover to generate succinct proofs of correctness for computations performed over large datasets, ensuring transparency and scalability.
 - b. **Polynomial Commitments:** These are cryptographic constructs used to commit to polynomials without revealing their coefficients. Polynomial commitments are essential for ensuring the integrity of computations in zk-STARKs and enabling efficient verification of proofs.
 - c. **Low-Degree Testing:** This is a technique employed in zk-STARKs to verify that polynomials have low degree, ensuring that they accurately represent the computations being proved. Low-degree testing enables the security and reliability of zk-STARK constructions.
- 3. **Bulletproofs:** Are a non-interactive zero-knowledge proof system that enables efficient range proofs, which verify that a number lies within a specific range without disclosing the number itself. Bulletproofs offer significant improvements in efficiency and scalability compared to previous range proof techniques. The key cryptographic concepts used in Bulletproofs are:¹⁷
 - a. **Inner Product Proof:** This is a cryptographic proof used in bulletproofs to verify that two vectors have a specific inner product. Inner product proofs enable efficient range proofs, allowing verifiers to confirm that a number lies within a predetermined range without revealing the number itself.
 - b. **Pedersen Commitments:** These are commitments to values based on the discrete logarithm problem in elliptic curve cryptography. Pedersen commitments are utilized in bulletproofs to commit to values while preserving their confidentiality. They facilitate the construction of succinct and efficient range proofs.
 - c. **Range Proof Verification:** This involves verifying that a committed value lies within a specified range. Range proof verification is crucial for ensuring the integrity of transactions in cryptocurrencies and other applications requiring confidential value transfers. Bulletproofs offer significant improvements in the efficiency and scalability of range proof verification compared to previous techniques.

By understanding these cryptographic primitives and their significance, readers can better understand the technical details of zk-SNARKs, zk-STARKs, and bulletproofs more effectively, thereby enhancing their comprehension of the underlying cryptographic protocols and their applications in privacy-preserving authentication.

3.4 | Evaluating literature

In this section, we outline the process of evaluating the literature included in Section 3.2. Initially, the literature was categorized broadly, with each category representing a specific use case for the applications included. This categorization aids in organizing the findings within the results and discussion section, linking them to their respective use cases and illustrated in Figure 8. It's worth noting that this categorization is not always precise, as some research works could fit into multiple categories. However, each piece of research is placed in the category we deemed most fitting for the presented solution.

Each category of literature was evaluated based on the methodology described in the following sections, and the findings are summarized in Table 6. The table includes the following abbreviations:

- 1. **Ref.**—Reference; Details and source of the work
- 2. **Application**—The application for which the research work proposed a solution using zero-knowledge proofs
- 3. **Use case**—Our use case grouping of the application
- 4. **Protocol**—The non-interactive zero-knowledge proof protocol used in the proposed solution
- 5. **Analyses**—The types of analyses performed in the work
 - a. **S**—Security analysis

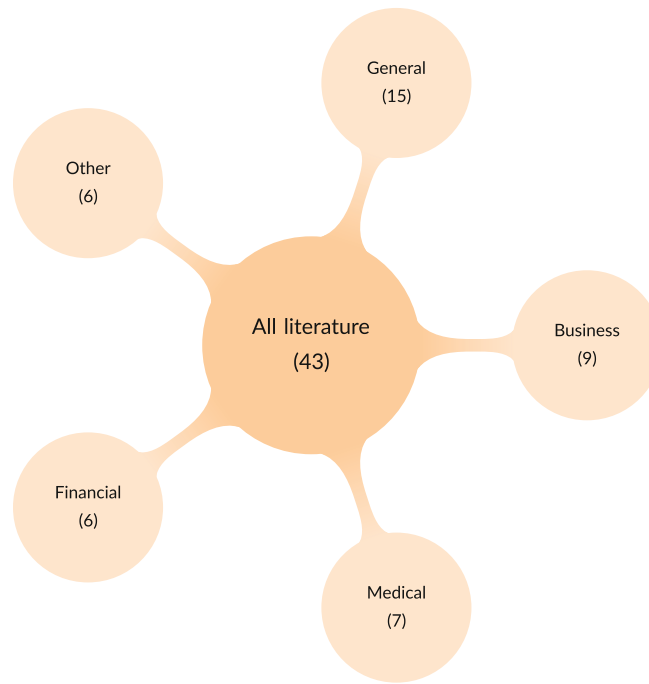


FIGURE 8 Research use case categorization.

- b. **SC**–Security analysis, including a comparison with other works
- c. **P**–Performance analysis
- d. **PC**–Performance analysis, including a comparison with other works
- e. **C**–Cost analysis
- f. **CC**–Cost analysis, including a comparison with other works
- 6. **Strengths & weaknesses**–Our findings on the strengths and weaknesses of the work

3.5 | Financial

In this section, we thoroughly examine and assess various research papers within the Financial category. zk-STARKs, zk-SNARKs, and Bulletproofs can be used in financial transactions to ensure privacy, transparency, and efficiency. For example, zk-SNARKs can enable anonymous transactions in cryptocurrencies like Zcash, while zk-STARKs can provide transparent and scalable verification of transaction validity. Bulletproofs can enhance the efficiency of range proofs, reducing the computational overhead of verifying transactions with confidential amounts. In financial use cases, zk-STARKs offer transparent and scalable zero-knowledge proofs without a trusted setup, ensuring efficient and auditable financial transactions, albeit with potential computational overhead. zk-SNARKs provide succinct and efficient proofs, enhancing privacy in financial transactions, but their reliance on a trusted setup may introduce vulnerabilities. Bulletproofs offer efficient range proofs, suitable for confidential transactions, though integrating them into financial systems may require adjustments. Notably, ZKPs have been instrumental in enhancing privacy and scalability in various blockchain networks. For instance, cryptocurrencies like Zcash and Monero utilize ZKPs to enable confidential transactions, allowing users to transfer funds without revealing transaction details such as sender, receiver, and transaction amount. Additionally, zkRollups, a layer 2 scaling solution, leverages ZKPs to aggregate multiple transactions into a single proof, thereby significantly reducing on-chain data storage and transaction fees on platforms like Ethereum. Furthermore, Ethereum 2.0's Beacon Chain implements ZKPs to enhance network security and efficiency, particularly in the context of shard chains and validator rotations. These examples underscore the diverse applications of ZKPs in blockchain technology, highlighting their pivotal role in addressing privacy, scalability, and security challenges in decentralized systems.

TABLE 6 Benchmark of the systematic literature review.

References	Application	Use case	Protocol	Analyses	Strengths & weaknesses
18	Digital currency	Financial	Bulletproofs	S, PC	+ Full implementation included and made publicly available + Quantum resistance briefly mentioned in discussing commitments – Limitations not mentioned – Future research directions not mentioned
21	Inter-bank settlements	Financial	zk-SNARK	C	+ Limitations mentioned + Future research directions mentioned – Experimental implementation not included
22	Asset exchange	Financial	zk-SNARK	S, P	– Quantum resistance not mentioned + Test implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
23	Transaction processing	Financial	zk-SNARK	P	+ Prototype implementation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned
24	Digital currency	Financial	zk-SNARK	S, P	+ Experimental implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
25	Asset exchange	Financial	zk-SNARK	S, P	+ Experimental implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
26	Medical record sharing	Medical	zk-SNARK	P	+ Simulated test implementation included + Future research direction mentioned – Limitations not mentioned – Quantum resistance not mentioned
27	Medical research data sharing	Medical	zk-SNARK	-	+ Future research direction mentioned – Non-theoretical implementation not included – Limitations not mentioned – Quantum resistance not mentioned
28	Medical record sharing	Medical	zk-SNARK	S, PC	+ Test implementation included + Future research direction mentioned – Limitations not mentioned – Quantum resistance not mentioned

(Continues)

TABLE 6 (Continued)

References	Application	Use case	Protocol	Analyses	Strengths & weaknesses
30	Business transaction tracing	Medical	Bulletproofs	S, P	+ Test implementation included and made publicly available + Limitations mentioned – Future research directions not mentioned – Quantum resistance not mentioned
31	Contact tracing	Medical	Bulletproofs	S, P	+ Test implementation included + Limitations mentioned + Future research direction mentioned – Quantum resistance not mentioned
32	Health monitoring	Medical	zk-SNARK	S, P, C	+ Test implementation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned
34	Medical insurance claiming	Medical	zk-SNARK	S, PC, C	+ Test implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
35	Record keeping & auditing	Business	zk-SNARK	—	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
36	Transaction auditing	Business	Bulletproofs	SC	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
39	Liability auditing	Business	Bulletproofs	S, P	+ Experimental implementation included + Limitations mentioned + Future research direction mentioned + “Post quantum friendliness” very briefly mentioned
41	Record keeping & auditing	Business	Bulletproofs	—	+ Future research direction mentioned – Non-theoretical implementation not included – Limitations not mentioned – Quantum resistance not mentioned

(Continues)

TABLE 6 (Continued)

References	Application	Use case	Protocol	Analyses	Strengths & weaknesses
42	Supply chain tracing	Business	zk-SNARK	S, C	+ Test implementation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned
43	Property sharing contracts	Business	zk-SNARK	S, P	+ Future research directions mentioned – Non-theoretical implementation not included – Limitations not mentioned – Quantum resistance not mentioned
44	Data auditing	Business	Bulletproofs	S, P	+ Test implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
45	Sealed-bid auctions	Business	Bulletproofs	S, P, C	+ Experimental implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
47	Attribute-based access control	General	zk-SNARK	S, P, C	+ Proof of concept implementation included + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned
48	Reputation system	General	zk-SNARK	S, P	+ Test implementation included + Future research direction mentioned – Limitations not mentioned – Quantum resistance not mentioned
49	Consensus system	General	zk-SNARK	—	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
50	Identity exchange	General	zk-SNARK	S, P	+ Test implementation of main building blocks included + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned

(Continues)

TABLE 6 (Continued)

References	Application	Use case	Protocol	Analyses	Strengths & weaknesses
51	Biometric authentication	General	zk-SNARK	S, P	+ Experimental implementation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned
58	Voting	General	zk-SNARK	S	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
52	Attribute-based access control	General	zk-SNARK	SC	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
59	Authorization system	General	zk-SNARK	S, CC	+ Test implementation included + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned
62	Location-based incentives	General	zk-SNARK	S, PC	+ Implementation simulation included + Future research direction mentioned – Limitations not mentioned – Quantum resistance not mentioned
65	Reputation system	General	Bulletproofs	S, P	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
66	Identity management	General	zk-SNARK	S, P, C	+ Implementation simulation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned
67	Federated learning	General	zk-SNARK	P, C	+ Experimental implementation included + Limitations mentioned + Future research directions mentioned – Quantum resistance not mentioned

(Continues)

TABLE 6 (Continued)

References	Application	Use case	Protocol	Analyses	Strengths & weaknesses
68	Biometric authentication	General	zk-SNARK	PC	+ Experimental implementation included + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned
69	Data federation	General	zk-SNARK	S, PC	+ Experimental implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
71	Vehicular data sharing	Other	zk-SNARK	S, PC	+ Proof of concept implementation included and made publicly available + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned
72	Vehicle-to-grid exchange	Other	zk-SNARK	S	– Non-theoretical implementation not included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
73	Vehicular digital forensics	Other	Bulletproofs	SC, PC, C	+ Test implementation included + Limitation mentioned – Future research directions not mentioned – Quantum resistance not mentioned
78	Tor abuse detection	Other	zk-SNARK	S, P	+ Test implementation included + Limitations mentioned – Future research directions not mentioned – Quantum resistance not mentioned
79	Vehicle-to-grid exchange	Other	zk-SNARK	S, P, C	+ Prototype implementation included – Limitations not mentioned – Future research directions not mentioned – Quantum resistance not mentioned
80	Mobile crowd sensing	Other	zk-SNARK	SC, PC, C	+ Experimental implementation included + Future research directions mentioned – Limitations not mentioned – Quantum resistance not mentioned

1. In Reference 18 the authors proposed PGC (pretty good confidentiality), a standalone cryptocurrency based on a decentralized confidential payment (DCP) system. PGC offers transaction anonymity and confidentiality, while conforming to regulatory compliance requirements by adding auditability. A new public-key encryption called twisted ElGamal is used to add zero-knowledge proofs like Sigma protocols and bulletproofs to prove transaction correctness while preserving privacy. The generic DCP system uses an integrated signature and encryption (ISE) scheme and non-interactive zero-knowledge proof (NIZKP) systems. Sigma protocols prove linear relations, while bulletproofs prove value intervals. The twisted ElGamal encryption scheme enables the use of both protocols.

The authors included a brief performance analysis that included proof size and transaction times for the generation and verification of different policy types in PGC. However, the authors also referred to a full version of their work which included an exceptionally extensive benchmark of the different parts of PGC. This full version also included a comparison of the proposed scheme against two other account-based DCPs, zkLedger¹⁹ and Zether.²⁰ The full version of this work also included a security analysis of the DCP construction which described the correctness of the generic DCP by first listing a theorem and then proving it. The authors did not conclude the paper with specific benefits or limitations of PGC, and neither did they provide any future directions. However, from the other content in their work, we could conclude that the main benefits of GCP were the security, privacy, and auditability aspects of the scheme.

2. In Reference 21 the authors propose a decentralized netting protocol for inter-bank payments, addressing the challenges of traditional RTGS systems due to single point of failure and trust issues. The protocol, combined with ElGamal encryption and zero-knowledge proofs, ensures settlement correctness, prevents permanent gridlocks, and preserves user privacy, thereby enhancing the security of inter-bank payments. The proposal uses ElGamal encryption to encrypt payment instructions, requiring the correct private key for successful decryption. Participants use zero-knowledge proofs like zk-SNARKs to verify submitted instructions and account balances. Blockchain technology, specifically smart contracts, is used for verifications, Merkle trees calculation, and account balance updates. Participants can prove knowledge of incoming payment instructions without disclosing transaction details. The authors conducted a cost analysis of Ethereum Gas for submitting and updating transactions on the Ethereum "Istanbul" fork, indicating that while the protocol is practical for deployment on Ethereum, it may perform better on permissioned blockchains.

While the main benefit of the proposed protocol was that it could resolve gridlocks in inter-bank payments while being decentralized, confidential, and privacy-preserving, the authors did mention some limitations to the protocol. These limitations included the need for all banks to be online for the settlement and a privacy leak in the protocol that could worst-case be exploited to successfully break anonymity. Resolving the latter limitation required modifications to the zero-knowledge proof in the update transaction, which the authors left for future works. Other suggested future research directions included investigating better proving times and a transparent setup method.

3. In Reference 22 the authors propose zkChain, a privacy-preserving model for efficient asset transfer using zk-SNARKs and one-way hash chains. They aim to reduce the proof size and verification times while ensuring the anonymity and privacy of participants. zkChain expresses account balances using one-way hash chains, with the last hash serving as public input and the preimage as private input. Provers can use zero-knowledge proofs to prove ownership, and the blockchain records account balance proofs for traceability.

An included performance analysis existed of two parts. In the first part, the authors compared the key generation, proof generation, and verification times for two different machines and listed the proving key and proof sizes. Regarding the results, the authors explained that the slower proof generation was not a problem for the system since only a single user performed it. While each participant had to perform the proof verifications, these verifications were much faster to perform than generating said proof. The transaction latency and send rate for an increasing number of participating nodes, along with the transaction delay for an increasing send rate, were displayed in the second part of the performance analysis. The outcomes demonstrated that while a higher send rate had no effect on performance, a higher node count had a negative effect. Additionally, the authors included a security analysis that analysed the privacy and attack resistance aspects of zkChain. This analysis described how zkChain guaranteed the zero-knowledge balance and transaction linkability aspects while being resistant to double-spending and over-spending attacks. In the end, the authors concluded that the performance of zkChain was its main competitive advantage compared to other solutions, while the techniques used in zkChain simultaneously prevented overspending attacks.

4. In Reference 23 the authors proposed a blockchain-based transaction processing system (TPS) to improve business communication and trading efficiency. However, they faced concerns about data confidentiality. They proposed a

solution using homomorphic encryption and zero-knowledge proofs, ensuring only authorized entities could view private transaction data, addressing the main concern in previous works. The proposal used homomorphic encryption to encrypt transaction information in the Bb-TPS to guarantee confidentiality. As a result of using homomorphic encryption, the system could still perform mathematical operations, like calculating the new account balance, on the encrypted transactions while the encryption hid the transaction contents. However, since the transactions were encrypted, it was hard to provide transparency as well. To this end, the proposal used zk-SNARKs as zero-knowledge proofs to prove and verify statements on the transaction without revealing any transaction details. These proofs provided the transparency required in the proposal. The accompanying performance analysis, which displayed the computation time and record size for a growing number of completed transactions, assessed the effectiveness of a prototype implementation. For the same quantity of transactions, the writers also included a performance summary for a SQLite database. However, no comparison with other Bb-TPS systems was done.

In conclusion, the main benefit of the proposed system was tamper resistance, data confidentiality, and improvements in accounting and auditing efficiency and effectiveness. These benefits however came with the downside of the computational overhead imposed by the blockchain. Other limitations of the current proposal included that no details were provided on the block mining and reward mechanisms, nor on the zk-SNARK implementation. The authors finally mentioned the development of blockchain-based continuous monitoring systems as a promising future research direction.

5. In Reference 24 the authors proposed Platypus, a central bank digital currency (CBDC) designed to offer privacy, scalability, and regulation. It combines e-cash transaction processing with ledger-based fund management, providing privacy guarantees and simple regulatory enforcement. Platypus' CBDC trust model with a central authority eliminates privacy and performance limitations of previous designs. The main enabling technology is zk-SNARKs, which verify transaction validity by the central bank. The security analysis included in the paper analysed transaction integrity, transaction privacy, fund availability, and regulation integrity and privacy. For each of these requirements, the authors first stated a definition, then stated a claim based on that definition, and subsequently proved the claim.

Additionally, the authors conducted a performance analysis. This analysis listed the trusted setup, proving, and verification times, as well as the number of constraints and the transaction size. These aspects were included for the base transaction and transactions with additional regulations. The proving times of the scheme were three to four orders of magnitude higher than the verification times, while the trusted setup takes an order of magnitude longer than generating a proof. The exact results however differed based on the used devices and elliptic-curve algorithms. To conclude, the authors mentioned the main benefit of Platypus was that, given the trust model of CBDCs, Platypus was able to provide all the desirable CBDC features of privacy, performance, scalability, and regulation using an e-cash-like system.

6. In Reference 25 the authors propose PrivateEx, a privacy-preserving scheme for exchanging assets. Unlike centralized platforms, PrivateEx uses zero-knowledge proofs to ensure fairness and correctness without publicly exposing all exchange information. This approach addresses privacy challenges faced by centralized platforms that require trust and expose all exchange information. PrivateEx uses zero-knowledge proofs, specifically zk-SNARKs, to ensure fairness, correctness, and privacy in asset exchanges. These proofs are publicly shared and verified using blockchain technologies, which also store exchanged assets. A security analysis outlines the system requirements and how PrivateEx ensures these requirements.

Additionally, the authors included a performance analysis that evaluated the performance of different zk-SNARK circuit designs on the number of constraints, proving and verifying key sizes, and proof generation times. The authors additionally remarked the verification times to be constant in the number of constraints. The main benefits of PrivateEx were the fairness, and correctness guarantees of the scheme, combined with the preservation of privacy during asset exchanges. In the conclusion the authors briefly mentioned that using Bulletproofs could have improved performance, however they did not elaborate on this.

In the financial category we observe several commonalities between the individual works. Most importantly each proposed solution aims to improve the privacy for some aspect of a system. Financial details specifically are known to be sensitive information as they provide knowledge on the means an individual. Besides that, monetary transactions can leak heaps of information on political affiliations (e.g., through donations), interests, acquaintances, and even enable the current location of a user. For this reason it therefore appears reasonable to want to hide transaction information for all these kinds of information to remain confidentiality. What also became clear though is that such confidentiality

was already possible in existing systems, but that this option comes to the detriment of other system aspects including transparency and the ability to provide auditability for the conformity to regulatory requirements. This too is an issue very specific to the financial category, in that the financial sector has some of the strictest regulatory requirements. These requirements aim to prevent illegal behaviour such as tax evasion, money laundering, and corruption, just to name a few, and additionally provide government oversight with the intention to protect customers and economic systems. From the works in this section it became clear that ZKPs are seen as a possible solution to the existing problem of combining privacy, anonymity and confidentiality, with regulatory conformity, transparency and correctness, since they enable in several ways to prove statements on transactions and financial systems while keeping the sensitive details hidden.

We can equally observed some interesting difference between the goals of the works. Starting with the difference between the use of a centralized (in Reference 24) or decentralized system architecture (in References 18, 21, and 25). This shows that not only are ZKPs useful in decentralized system applications including ZCash and Monero as discussed in this section introduction, but that they can equally be useful in conforming to regulation while keeping sensitive details private in centralized systems including a central bank digital currency as proposed in Reference 24.

Finally we discuss an observation on how the authors of the works describe the impact of the ZKP protocols on the performance of their proposed solutions. In References 21 and 23 the authors express their interest in reducing the computation times, or the downside of the computational overhead of their proposal. Though for the latter work this computational overhead was attributed to the blockchain and not the use of ZKPs. In Reference 24 the authors expressed contentedness with their system attaining performance targets, while ... We ascribe this difference to authors adding ZKPs to an existing system idea to add some of the benefits of the ZKP, and then seeing the performance drop compared to that system, as opposed to authors devising an entire system using ZKPs from the start, with them being contend in the performance given the trust model and the included privacy and scalability.

3.6 | Medical

This section discusses and reviews case studies under the Medical category. In the medical field, zk-STARKs, zk-SNARKs, and bulletproofs can facilitate secure data sharing and ensure patient privacy. For instance, zk-SNARKs can be used to create verifiable medical records without revealing sensitive information, while zk-STARKs can enable efficient auditing of medical data across multiple parties. Zk-STARKs offer transparent and scalable zero-knowledge proofs without a trusted setup, enabling secure data sharing with verifiable integrity, although they may entail computational overhead. zk-SNARKs provide succinct proofs, enhancing privacy in medical records, but their reliance on a trusted setup may pose security risks. Bulletproofs offer efficient range proofs, ensuring the integrity of medical data with minimal computational overhead, though integrating them into existing medical systems may require adjustments and expertise. Overall, while each protocol offers unique strengths in ensuring privacy and integrity in medical applications, their limitations must be carefully considered in deployment to address potential challenges effectively.

1. In Reference 26 the authors propose a blockchain-based system for securely storing electronic health records (EHRs) to address privacy and security concerns. EHRs are widely used in health institutions for access to accurate information, but they can be hacked, limiting patient access. The proposed solution ensures privacy through zero-knowledge proofs, adding immutability, authenticity, and accountability. The protocol uses the zk-SNARK zero-knowledge proof protocol for authentication, enabling patients to verify their doctor's medical license, insurance, and prescription. This method maintains privacy-sensitive data without revealing the data. Smart contracts are used for functions like adding doctors, inserting EHRs, viewing EHRs, and sending prescriptions. These actions are recorded on the blockchain as events. The system combines the interplanetary file system (IPFS) with blockchain to store encrypted EHRs in a distributed manner.

The authors evaluated the system performance of the proposed system using a simulated local blockchain called Ganache, while they used a "calliper" to record the performance of the smart contract implementations of Hyperledger Sawtooth and Ethereum. The resulting performance metrics, which demonstrated the latency and throughput figures for different send rates, were included in a performance analysis. Besides the performance figures of the smart contract implementations, however, the authors did not analyse other performance aspects or the security of the system. In conclusion, the authors mentioned that the main benefit of the proposal was that patients had unrestricted access to their EHRs, while the system guaranteed the privacy and consistency of the EHRs. The authors finally

suggested research into the application of the proposed paradigm in areas besides healthcare as a future research direction.

2. In Reference 27 the authors propose a privacy-preserving data-market platform for medical data, particularly for machine learning and AI research. They use zero-knowledge proofs and blockchain technologies to ensure privacy, verify data validity, and offer monetary incentives for data sharing, addressing patient resistance to data misuse and lack of transparency. The proposed platform uses blockchain technologies to deploy smart contracts that automatically update a machine learning model based on personal data contributions. These contracts also automatically remunerate participants for data contributions, protecting patient privacy. The platform uses zk-SNARKs for zero-knowledge proofs to validate the accuracy of contributed parameters. The smart contracts update the model on the distributed ledger only if the validation succeeds.

The authors did not include a performance or security analysis according to our definition; however, they instead provided a detailed mathematical description of the quadratic span problem in the zk-SNARKs. In conclusion, the authors mentioned that the main benefit of the proposed solution was the automated way researchers could update their machine-learning models with verified contributed patient data. The ability to do this using real data was partially a result of the main benefit for patients, namely privacy protection and remuneration, which increased the willingness of patients to contribute. The authors finally briefly mentioned the research into the further applicability of blockchain technology as a future research direction but without more details.

3. In Reference 28 the authors propose a privacy-preserving blockchain-based scheme for securely sharing medical data between patients and research institutions. They argue that existing systems struggle with patient reluctance and the need for certainty in the usefulness of the information. They propose a solution that combines blockchain technologies with zero-knowledge proofs, aiming to address issues like anonymous secure authentication and uncertainty in the applicability and correctness of received information. This approach aims to create a secure and private medical data-sharing system. The proposed scheme uses zk-SNARKs to generate proofs of data requirements without disclosing them, and for patients and health institutions to verify submitted data conformity. A blockchain smart contract verifies these proofs, and if successful, encrypted data is uploaded to a semi-trusted server for decryption. The transaction is verified and published on the blockchain for finalization and recording. The research conducted a security analysis on the proposed solution, highlighting its ability to achieve confidentiality, availability, integrity, privacy preservation, and traceability, while also preventing single points of failure.

Next to the security analysis, the research included a performance analysis evaluated on a test implementation. This analysis listed for different numbers of circuit inputs the key generation, proof, and verification times, as well as the proving key, verification key, and proof sizes. The performance analysis additionally compared these performance factors against PGHR,²⁹ which showed favorable results for the proposed solution in the key generating keys and proving phases but unfavorable results for verifications.

The researchers compared a proposed scheme to three other ones on computation costs, startup nodes, and privacy protection. They concluded that the main benefits were privacy protection and data availability and consistency. They suggested future research should focus on optimizing implementation efficiency and ensuring the scheme can resist conspiracy attacks.

4. In Reference 30 the authors propose a secure protocol for genomic range queries between users and testing facilities, aiming to improve privacy and efficiency while preserving security. They use cryptographic commitments, zero-knowledge range proofs, and homomorphic encryption to achieve their desired functionality. Cryptographic commitments involve committing the positions of genome mutations at the sequencing lab, while zero-knowledge range proofs use bulletproofs to prove the positions of the first two mutations outside the queried range. Additionally, homomorphic ElGamal (AH-ElGamal) is used as the most efficient homomorphic encryption scheme for similarity testing, allowing users to decrypt the total value to calculate the similarity score. This approach addresses the tension between privacy and security in genomic range queries.

The authors evaluated the proposed protocol through performance and security analyses, comparing it to alternative homomorphic encryption schemes AH-ElGamal and Paillier. They also assessed the system's authenticity, completeness, and user privacy requirements, providing proof of how the implementation met these requirements. Some of the limitations of the system included that the used additive homomorphic encryption did not support all types of genomic tests and that the protocol only supported the single-nucleotide polymorphism (SNP) genomic representation. Furthermore, the authors avoided discussing low-level security problems such as side-channel attacks.

Contrarily, the main benefit of the proposed protocol was that it was an efficient protocol that accounted for security and privacy requirements. While the proposed protocol focused on revealing plaintext SNP representations of genomes, the protocol was mentioned to be equally applicable to encrypted genomes.

5. In Reference 31 the authors propose a blockchain-based privacy-preserving contact tracing system for tracing infectious diseases. Traditionally, in-person interviews were unreliable and time-consuming. Smartphones enable automated contact tracing using location-based or contact-based methods, but these are privacy-sensitive and reduce user privacy. The proposed system aims to solve these issues by generating zero-knowledge range-proofs to check user contact without revealing their precise locations. The proposed system uses bulletproof zero-knowledge range proofs to prove a user's location range, hashed and stored on the blockchain. Users who are confirmed infected disclose their location range during a specific period. Other users can verify their location ranges using verified ranges. Users without overlaps can submit new range proofs. Blockchain addresses are used as substitutes for real user identities to hide their identities. The proposed system's security analysis outlines assumptions and guarantees for location privacy, anonymous identification, verification information exposure, and false positives. Performance analysis of a test implementation on Hyperledger Fabric reveals that response times increase linearly with the number of apps (users), highlighting the system's security measures.

The system's performance could have been improved by ensuring user privacy and addressing performance issues with large users. Additionally, the system could not verify the validity of location data from sensors on user devices. The authors suggest researching ways to verify this data validity as a future research direction. Despite these limitations, the proposed solution hid user locations and allowed users to prove their distance from infected users using their footprints.

6. In Reference 32 the authors propose a privacy-preserving, blockchain-based healthcare system for IoT devices, ensuring secure and private sharing of health data. While cryptographic techniques can implement security, privacy is difficult without a trusted third party. Blockchain-based systems remove the need for a trusted third party and add anonymity, but their public nature introduces new privacy challenges. Zero-knowledge proofs solve these problems, creating private and secure healthcare systems on the blockchain. The proposal solution uses blockchain technology, including smart contracts, to provide a distributed platform with strict data integrity guarantees. Blockchain addresses act as pseudonyms, while smart contracts run computations and verify zero-knowledge proofs, such as zk-SNARKs, for authentication to the HSP and IoT devices. Common cryptographic techniques like digital signatures, hash functions, and Diffie-Hellman key exchange are used. The proposed system met user anonymity, authentication, and confidentiality requirements through a security analysis, which was demonstrated through security theorems and security games. Furthermore, The performance analysis of the system using the Ropsten³³ test-net blockchain network evaluated the initialization, proof, and verification of zk-SNARKs for various user numbers. The authors also conducted a cost analysis of interactions with the smart contract, revealing Ethereum gas costs and conversions.

In conclusion, the main benefit of the proposed system was that it enabled the secure and private sharing of health data between IoT devices and health service providers, without the need for third parties. The authors did mention some limitations of the current system, including the high computational power requirement and the high costs for device management. To minimize these downsides, the authors proposed research into lower computational power requirements and the usage of non-EMV blockchain networks as future research directions.

7. In Reference 34 the authors propose a privacy-preserving medical insurance claim scheme using blockchain technology and zero-knowledge proofs, addressing existing complex, unreliable, and data leaky schemes and addressing fraud concerns in medical insurance purchasing and claiming scenarios. The proposal aims to implement a smart contract system for insurance purchases using blockchain technologies, zk-SNARK zero-knowledge proofs, Schnorr protocol, Fiat-Shamir heuristic method, and homomorphic encryption based on Decisional Bilinear Diffie-Hellman (DBDH). These technologies verify the identity and medical data of patients, initiate insurance claims, generate new medical data, and inform the insurance company to transfer money to the patient. A security analysis of the proposed scheme is also included, providing a security definition and proof for both. The work additionally included a performance analysis of the proposed system, which evaluated the performance using a test implementation of the system on a locally running private Ethereum test network. This analysis showed the proof generation and verification times for increasing numbers of inputs, message sizes, and ID bit sizes. Also included in the performance analysis was a brief comparison to ADSNARK,²⁹ which showed that the proposed solution had comparatively lower proof generation times. The higher verification times of the proposal on the other hand were justified by the authors because the proposal performed additional verifications not included in ADSNARK.

Finally, a brief cost analysis showed the Ethereum transaction fees in gas for deposits, proof requests, and verification requests. In conclusion, the main benefit of the proposed solution was that it included the properties of authentication correctness, completeness, and perfect zero-knowledge. Through these properties, the system protected patients' privacy and guaranteed the legitimacy of user identities.

In the medical category we observed a very clear trend. Each of the in the works proposed solutions aims to improve the privacy for people by hiding the direct medical data from other people and institutions. This makes sense, medical data directly relates to someone's well-being and personhood, up to the most intimate aspects. This makes medical data very privacy sensitive data when expose unwillingly. It also became very clear however that it can be of dire importance to share such medical data with medical institutions, be it for doctors to have access to EHRs as in Reference 26 or data from IoT devices as in Reference 32, to detect health problems early by testing for genetic mutations as in Reference 30, or for research institutions to learn and create cures for medical conditions as in References 27 or 28. By using ZKPs to enable proving certain statement on the privacy sensitive medical data, this can prevent the need to directly expose the raw medical data if proof of the statement itself is enough to obtain the required knowledge. A remaining work that can have a direct impact on the health of a society³¹ does not use ZKPs to hide medical data, but to hide location data instead. The proof on such location history provides a way to allow contact tracing without leaking this location history. When the location matches that of an infectious person, they can be informed to isolate to thereby prevent the further spread. The remaining work,³⁴ while not directly or indirectly beneficial for the medical health of the involved person, is important for patients to prove a medical claim to their insurer without having to provide the exact details of the medical care. It is enough for the insurer have proof that the patient required and received this care to process a payout for said treatment. This again makes it so the patient does not have to share the medical details, where without the use of ZKPs they would not be able to receive a payout without providing this sensitive data.

Another pattern we observed for the medical category is that the authors of each work are less interested in really improving the performance, just Reference 32, but more so in improving the security, as in Reference 28, or the application of their solution to different situations and paradigms, as in References 26 and 27. It quickly becomes clear why this is the case when taking into account that personal health is often considered the most important thing in life. In the medical sector it is therefore more logical to think about the safety and applicability of a solution, before considering the costs of applying said solution.

3.7 | Business

Businesses can leverage zk-STARKs, zk-SNARKs, and bulletproofs to enhance data privacy, auditability, and trust in transactions. For example, zk-SNARKs can enable confidential transactions in supply chain management, protecting sensitive information while ensuring transparency. zk-STARKs can provide verifiable proofs of ownership or authenticity in intellectual property protection, while bulletproofs can improve auditing processes by providing efficient range proofs. In business applications, zk-STARKs offer transparent and scalable zero-knowledge proofs without a trusted setup, ensuring trustless and auditable transactions, although they may have higher computational requirements. zk-SNARKs provide succinct proofs, enhancing privacy and auditability in business transactions, but their reliance on a trusted setup may introduce vulnerabilities. Bulletproofs offer efficient range proofs, ensuring confidentiality and integrity in business transactions with minimal computational overhead, although integrating them into existing systems may require adjustments and expertise. While each protocol offers unique strengths in enhancing privacy and trust in business operations, their limitations must be carefully addressed to ensure seamless integration and effective deployment.

1. In Reference 35 the authors propose a record-keeping infrastructure that uses blockchain technology, zk-SNARKs, and commitments to achieve privacy, public auditability of secret data, accountable deletion, and succinctness. This solution addresses the issue of encryption preventing governance and ensuring transparency and accountability in government agencies. The infrastructure preserves privacy by encrypting data and committing to them, which can be opened to prove compliance. The commitments are recorded on the blockchain, making them public and immutable. Zero-knowledge proofs, specifically zk-SNARKs, are used to prove compliance with secret regulations. A commitment is created for each secret regulation, which is then used to prove compliance using the secret regulation commitments. Finally, the authors discussed several ways to utilize or reduce the fees involved in adding records to the blockchain.

One of the considered options was to let miners guarantee the correctness of the zk-SNARK in the record before it was appended. In this case, a fee had to be appended to the record for each regulation the record did not comply with, which would act as a penalty for not guaranteeing compliance with those regulations. We inferred from the contents of the proposal that the main benefit was that the system was the first auditable record-keeping system that ensured compliance with secret regulations while preserving the privacy of records in the system.

2. In Reference 36 the authors propose AMLChain, a transaction confidentiality and integrity-preserving system for institutions implementing anti-money laundering and know-your-customer practices. This system uses blockchain-based distributed identities and security techniques, including zero-knowledge proofs and digital signatures, to improve the effectiveness of these practices while preserving user privacy, thereby reducing the cost and incompatibility of existing AML solutions. The proposal uses ElGamal asymmetric encryption for transaction signing and verification, uploading information to a public ledger using blockchain technologies. Pedersen commitments hide transaction amounts, while bulletproofs provide zero-knowledge range proof for transaction validity. These proofs help identify suspicious transactions and ensure funds remain within a certain value range. The included security analysis showed in five lemmas³⁷ that AMLChain was secure and private when the security assumptions of the used technologies held. Also included in the security analysis was a comparison to several other schemes that compared the sender, receiver, and transaction privacy aspects as well as the efficiency and the support for transaction graphs, transaction identification, and privacy audits.

While the comparison to other schemes included the scheme efficiency, the performance of the proposed scheme was not analyzed but only briefly discussed. In this discussion AMLChain was mentioned to perform comparably to FabZK,³⁸ which itself had a throughput several orders of magnitude higher than zkLedger.¹⁹ In conclusion, the main benefit of AMLChain compared to alternatives was that AMLChain included several privacy features and showed improved efficiency.

3. In Reference 39 the authors propose DAPOL+, a proof of liability (PoL) scheme that extends the existing DAPOL (distributed auditing proofs of liabilities) protocol to provide provable privacy and security. Traditional liability-proving schemes lack correctness guarantees and allow auditors to learn sensitive information. DAPOL+ adds formal PoL definitions and security proofs to prove its strong privacy and security, improving the overall security of the scheme. The proposal employs sparse Merkle trees (SMTs) alongside homomorphic Pedersen commitments and bulletproofs as zero-knowledge range proofs. In DAPOL+, the prover commits total liabilities, each user confirms their amount. The SMT accumulates liabilities, allowing the prover to prove total liability by revealing the root node's blinding factor. Bulletproofs ensure validity without overflow. The paper highlights the security of DAPOL+ and provides a separate chapter analyzing its failure probability, focusing on the possibility of the prover manipulating or discarding individual liabilities without detection. The authors developed a proof-of-concept implementation of DAPOL+ and conducted a performance analysis, revealing proving times, verification times, and proof sizes for various aggregated ranges. They suggested optimizations like trading larger proof sizes for faster proof generation.

According to the authors, the main benefit of DAPOL+ was that it implemented a PoL protocol with provable security and privacy. While a downside of DAPOL+ was that discarded liabilities of users that never verified remained undetected, the authors thoroughly discuss the probability of a system failure caused by such alterations. The authors finally proposed the formal treatment of additional features for particular applications as a future research direction.

4. In Reference 40 the authors proposed the usage of zero-knowledge proofs for corporate networks. Private transactions on the blockchain remained an open security concern, even though certain businesses had previously used blockchain technologies to give transactions traceability and to do away with the requirement for a trusted third party. Existing attempts to provide private transactions had limitations in decentralized storing and immutability verifications, both key factors of blockchain. By using zero-knowledge proofs the authors attempted to solve the limitations of private transactions on the blockchain for corporate use. However, the research only shallowly discussed the application of zero-knowledge proofs and otherwise described how previous implementations worked. Given that the authors kept the discussions on the level of a general idea and that the contents did not describe details of a specific application, we should not have included this research according to our filtering rules. Consequently, we have retroactively removed it from our analysis.
5. In Reference 41 the authors proposed an advanced zero-knowledge ledger system that was based on zkLedger.¹⁹ Traditional blockchain-based ledgers showed privacy issues, which the previous work zkLedger solved using zero-knowledge proofs. However, the main downside of the zero-knowledge range-proofs used in zkLedger was the

resulting slow performance. By replacing the range-proof in the existing zkLedger system, the authors of this proposal made the proof and verification times of zkLedger five times faster and enabled further efficiency improvements by aggregating multiple range-proofs into one. The main enabling technology in the proposal was the bulletproof zero-knowledge proof protocol, which replaced the previous range-proof scheme in zkLedger based on Borromean ring signatures. Provers used the bulletproofs to generate proof that their secret data value lay within a certain range without revealing the value. The prover then submitted the proof to the verifier for verification. The main benefit of the proposed solution was the improved system efficiency achieved through the reduction of the proof size using Bulletproofs. This improvement lowered the required memory and computational power requirements, which was especially useful for resource-constrained devices such as sensors and mobile devices. The authors finally suggested the actual implementation of the proposed improvement in the zkLedger prototype as a research direction for future work.

6. In Reference 42 the authors propose a privacy-preserving, traceable supply chain system on the public permissionless blockchain (PPBC), which improves existing PPBC-based systems by concealing private information and preserving privacy. This solution, using encryption and zero-knowledge proofs, addresses the issue of public readable private information in PPBC-based systems.

The proposal uses elliptic curve ElGamal encryption to encrypt the owner's blockchain address using the manufacturer's public key, allowing tracking of products. Zero-knowledge proofs, zk-SNARKs, are used to ensure product distribution without leaking information. A secret token is shared between the owner and recipient, ensuring correct distribution while keeping blockchain addresses hidden. Smart-contract blockchain technology manages manufacturer information and zero-knowledge proof verification.

Included in the research was a security analysis, in which the authors verified the traceability and privacy-preserving aspects of the proposal. To do so they first mentioned the possible attack vectors on traceability and privacy and then described why these attacks would fail. One of the attacks, a collusion attack between the owner and recipient, was still able to succeed. The authors proposed to investigate countermeasures to this attack in future research. Additionally, the authors conducted an evaluation of a test implementation on the Ethereum blockchain. Using this evaluation, a cost analysis was included that showed the Ethereum gas costs for different steps in the system for manufacturers and owners/recipients.

While the main benefits of the systems included that the system achieved privacy preservation, enabled traceability, and prevented unauthorized distribution, there were also some downsides. The relatively high Ethereum gas cost was one, while another downside was that owners could be inferred from the execution of smart-contract execution history. That is, the proposal did not preserve privacy at a protocol level. To resolve this privacy problem the authors suggested the preservation of privacy at the protocol level as a future research direction. Two other suggested future research directions included a scope increase by reducing transaction fees, and a system extension to deal with product assembly and disassembly.

7. In Reference 43 the authors propose PRC, a privacy-respecting contract platform for blockchain-based sharing economy applications. They aim to improve privacy for involved parties while maintaining the desired features of blockchain-based systems. They use three main technologies: commitments, Zk-SNARKs, and blockchain technology. Commitments commit a time-slot reservation or payment with a random value to the blockchain, hiding the committed values. Zk-SNARKs prove a party knows the random number of valid commitments, enabling verification. Blockchain technology validates and records commitments in an immutable and publicly visible manner.

The use of blockchain technology prevented double usage, that is, renting a property to multiple users at the same time. Furthermore, proxy agents were used to execute the contracts and remove the connections to real identities. The work included security and performance analyses. In the security analysis, the authors described how double usage attacks were prevented and how the system guaranteed the privacy of users and property owners. The performance analysis existed in two parts, the first of which was a discussion on the blockchain in which the authors stated that the throughput of blockchain could be a problem for the demands of certain sharing economy applications. The second part showed a performance evaluation of the zk-SNARK scheme, which was the most computationally expensive primitive in the PRC platform. The evaluation showed the key generation, proof generation, and verification times, as well as the proof size, and was performed for a 128-bit security level using a 1-million-gate circuit and 1000-bit inputs.

While the main benefit of the proposal included that the system preserved the privacy of owners and users while functioning on the blockchain, there were also some downsides to the system. These downsides included the

fixed contract durations and pricing, as well as the missing support for a privacy-preserving rating and recommendation system. The authors finally proposed to remove one of the downsides of the proposed system by adding a rating and recommendation system as a future research direction. They furthermore proposed to include more operations in the system and to evaluate the possibility of recovering links to actual identities as other future research directions.

8. In Reference 44 the authors propose zkrcChain, a privacy-preserving data-auditing solution based on the Hyperledger Fabric consortium blockchain. Unlike existing solutions, zkrcChain keeps transactor identities public while hiding transaction details for privacy. This allows for arbitrary-range audits and joint audits on data from multiple parties, making it suitable for scenarios like food safety supervision. The main difference between arbitrary-range zero-knowledge proofs and standard-range ones is that arbitrary-range proofs prove values within a smaller range.

For the system design the authors used the Hyperledger Fabric and blockchain technologies such as smart contracts to perform distributed tasks including initialization, retrieving range limits, and performing verification of submitted proofs. Bulletproofs, the chosen zero-knowledge range proofs, were used to generate proofs that the underlying data was within a certain value range without revealing the value itself. Contrary to the “one-prover-one-verifier” mode, the “multi-prover-one-verifier” had a dealer peer aggregate proofs from multiple provers before the (combined) proof was uploaded to the public ledger.

The research included a security analysis that first described the threat model for zkrcChain, after which the security mechanisms in zkrcChain that prevented these threats were described. The authors conducted a performance analysis of zkrcChain, a system implemented on Hyperledger Fabric, comparing its execution time and on-chain storage space usage to regular bulletproofs on standard-range proof sizes. They noted that comparisons to alternative systems were not possible due to the lack of similar functionality in zkrcChain. The comparison to regular bulletproofs showed that zkrcChain produced larger proof sizes because of the inclusion of intermediate challenges and the usage of a less efficient library implementation. In conclusion, the main benefit of zkrcChain was the support for the generation and verification of arbitrary-range zero-knowledge range proofs, while another great benefit was the aggregation of multiple proofs into one for improved efficiency and system applicability.

9. In Reference 45 the authors propose a blockchain-based sealed-bid auction protocol to address issues like collusion, ring bidding, and information leakage. The proposed solution aims to be privacy-preserving and publicly verifiable, but faces challenges like high communication costs, making it difficult to apply to large-scale auctions with multiple bidders. The proposal aims to achieve privacy and public verifiability through Pedersen commitments, blockchain technology, and zero-knowledge proofs. Pedersen commitments enable participants to commit bids, while Bulletproofs prove the value within a certain range. Range proofs announce auction winners without leaking other information. Blockchain technology ensures transparent, publicly verifiable, and irreversible transactions without a trusted third party.

The included security analysis consisted of three parts. The first two parts were on correctness and privacy preservation, in which the authors described a theorem and subsequently provided proof of the theorem. The third part of the analysis described how the proposed scheme met the requirements of public verifiability, fairness, and non-repudiation of bidders and auctioneers. Additionally, the authors conducted a performance analysis for an off-chain implementation, focusing on proving and verifying times of bids with increasing bit-lengths and bid aggregation and win verification times for increasing bidding participants. They also conducted an Ethereum gas cost analysis for each number of bidding participants, finding the scheme’s cost acceptable from 512 participants. The practicality for use in real-world applications enabled through the reduced complexity and communication costs of the proposed solution, was also the main benefit of the proposed solution.

The general observed trend in the business category is to devise new technologies in order to improve the ability to perform business transactions. Very important in these transactions is the transparency and traceability, usually provided through some sort of blockchain, which enables a trust relation between business partners and with customers. As observed however just as important is the consideration of confidentiality of private transaction details or business sensitive information, as well as the integrity of the transactions. These properties are provided by using one of the ZKP protocols. In short, this allows businesses to perform business transactions in a transparent and traceable manner required to conform to regulatory and trust requirements, while also enabling liability, integrity, and auditing of these transactions in a manner that prevents the sensitive details from leaking.

Whereas the general trend is clear, the included works introduce relating solutions for applications that still very much different from each other. Works in References 35, 41, and 44 focus on a ledger like system, where records are permanently written which preserves integrity and enables transparency and auditability while keeping the private information hidden. Work³⁶ implements a similar transaction recording system to enable conformance to know-your-customer regulations in such a way that private customer details remain confidential. The authors of Reference 39 worked on a system to prove not regulatory conformity but liability towards their customers, where the institution can prove they have enough funds for the liabilities of all customers participating in the proof, without disclosing exact financial numbers. Finally two works, References 43 and 45 use ZKPs to add guarantees on honesty and availability in the process. In Reference 43 this honesty and availability is applied to contracts in a sharing economy, where customers reserve resources for a limited time in exchange for a financial remuneration. In Reference 45 on the other hand the goal is to verify the honesty and availability of payments in auction processes while keeping transaction details private. This is in contrast to traditional auctions where either the transaction amounts and bidder identities are disclosed, or the auction can be manipulated by dishonest auctioneers or bidders.

The generally observed benefits of applying ZKPs in new business solutions are mostly in enabling privacy and security in combination with compliance or other requirements, where traditionally this was not possible, as mentioned by the authors from References 35, 36, 39, 42, and 43. Otherwise the improved performance or scalability was attributed as the main benefit of the introduced system, including in works in References 36, 41, 44, and 45. It stands to reason why these two options are mentioned as the main benefits of the newly introduced solutions, since they open up new business ventures where previously either privacy requirements were not adequately satisfied or the solution was not scalable enough. Both of these would limit the reach of a solution.

In the light of business opportunities the observed tendency of suggesting expanding the proposed solutions, as in References 39, 42, and 43, or the further reduction of (computational) costs, as in References 35, 39, and 42, become understandable or even logical.

3.8 | General (multiple use cases)

In this section, we discuss multiple use cases that can fit into more than one category as suggested.

1. In Reference 46 the authors proposed a zero-knowledge proof system for use in cluster computing, with a focus on the MapReduce cluster computing technique. The authors designed the proposal to prove the correctness of an output computed by a cluster node, without leaking the used private input data. By ensuring the zero-knowledge proof itself was also a cluster computation, the proposal solved existing challenges in generating such proofs in parallel in a cluster computing setting. The proposal introduced a bootstrapping theorem that used zk-SNARKs to obtain multi-predicate proof-carrying data (PCD) systems. These PCD systems were subsequently used on compliance predicates to obtain distributed zk-SNARKs. Cluster computing systems could then compute the resulting distributed zk-SNARK using the MapReduce technique. Given however that the proposal focused on the bootstrapping techniques for generating distributed zk-SNARKs, it did not comply with our manual filtering criteria of proposing a direct, privacy-preserving application of zk-SNARKs. We therefore concluded that in hindsight we should not have included this research in the manual filtering step, and consequently, we removed it from this analysis.
2. In Reference 47 the authors proposed a new architecture integrating the self-sovereign identity (SSI) model into attribute-based access control (ABAC) systems. The introduced architecture implemented the Extensible Access Control Markup Language (XACML) standard and was proposed as an improvement to centralized ABAC systems where attribute managers managed and disclosed data without control for the user. To achieve their objectives the authors used zk-SNARKs in the proposal to prove the result of a function over an identity attribute, without disclosing the attribute information. The research analyzed the performance of the proving key and circuit, cost analysis, and security analysis of the proposed system. It revealed that deployment and evaluation operations were more expensive on the Ethereum blockchain compared to Polygon or MATIC chains. The study also highlighted potential attacks and their mitigation strategies, demonstrating the system's security.

Attribute confidentiality, user control, and XACML compliance were the main benefits of the proposed system. Finally, the authors mentioned future research directions which included making further improvements to the performance and compatibility of the proposed system, as well as researching setting attribute disclosure constraints that guarantee a desired level of privacy.

3. In Reference 48 the authors propose a privacy-preserving reputation scheme for collaborative systems using pseudonyms to allow private interaction. This prevents reputation reset by using a new pseudonym, addressing issues with malicious users removing bad reputations. The proposal also addresses user reluctance to give negative feedback due to fear of retaliation. By using reputation tokens with blockchain ledgers and zero-knowledge proofs, the scheme addresses these issues without a trusted third party and includes public verifiability.

The proposed solution uses Pedersen commitments, zk-SNARKs, and blockchain technology to create reputation tokens. The blockchain acts as an append-only ledger, ensuring reputation tokens are only added and not altered. Pedersen commitments create reputation tokens with a user's secret key, serial number, random, and unique identifier. Reputation updates involve creating new tokens with an incremented serial number and a zero-knowledge proof. Zk-SNARKs prove committed attributes and pseudonyms belong to the same person. The scheme also allows peers to prove lower reputation bounds instead of the actual reputation. The research involved a security and performance analysis, with the authors presenting security and privacy theorems and providing proofs. Further details can be found in the full version of the paper. In the performance analysis, the authors showed the performance of the zk-SNARKs for the mint and update statements respectively, which each listed the key generation times, proving and verification times, and proof sizes.

The authors proposed using a trusted platform module (TPM) to replace registration tokens and prevent Sybil attacks. They proposed two methods: bind identities to human beings, which is Sybil-proof but requires personal information access, or bind identities to devices, which is more private but allows attackers to create multiple identities. In conclusion, the main benefit of the system was that it ensured the privacy and forward anonymity of users in the reputation scheme, while it prevented Sybil attacks and other reputation-based attacks. A future research direction proposed by the authors was to research securely combining update transactions without posting these update transactions to the blockchain, which would help increase the efficiency of the scheme.

4. In Reference 49 the authors proposed a privacy-preserving version of the proof of stake (PoS) consensus protocol Ouroboros Praos. PoS protocols proved to be a promising alternative to the proof of work (PoW) consensus protocol that induced high computational power and energy usage. However, PoS protocols proposed in existing works came with the downside of disclosing the identity and wealth of the stakeholders. This made PoS protocols unsuitable for privacy-preserving cryptocurrencies. In this research, the authors proposed the notion of an "anonymous verifiable random function" (AVRF) to create a private proof of stake (PPoS) protocol. In short, hiding stakeholder wealth required identity anonymity since otherwise the "win" frequency would have leaked the wealth of that stakeholder.

The proposal utilized zero-knowledge proofs, sigma protocols, Pedersen commitments, and Merkle trees as main technologies. Participants in the PoS protocol committed a stake using Pedersen commitments. Sigma protocols were used to prove a stake won the lottery, and the winning stake owner had to prove ownership using Merkle trees. The stake used in the winning lottery proof corresponded to the user's signature and AVRF keys, revealing a valid path to claim membership. Zk-SNARK zero-knowledge proofs were generated to validate the path and membership without revealing the leaf node. The proposal did not provide a security analysis.

The authors furthermore did not include a performance analysis, limitations section, conclusions, or future research directions.

5. In Reference 50 the authors proposed PrivIdEx. PrivIdEx used blockchain and zero-knowledge proofs to enable private identity information exchange between service providers in a secure and privacy-preserving manner, all without a centralized broker. The proposed solution reduced the number of costly user verifications, currently performed by each service provider separately for the same user, through the reuse of verifications performed by other services. PrivIdEx used zk-SNARKs as zero-knowledge proofs to prove that an identity satisfied the verification requirements while guaranteeing privacy and security. The Hyperledger Fabric was then used to perform such verification without a central broker.

The authors included a security analysis for the protocol and the users' privacy, which included proof that the stated security and privacy requirements hold. Also included was a performance analysis that analyzed circuit size, runtime, and storage requirements for varying identity asset and nonce sizes. The authors listed different versions of PrivIdEx with incremental security and privacy guarantees, each using a different circuit. Performance results for the different circuits were included in the analysis. The proposed PrivIdEx system offers anonymity and unlinkability guarantees, ensuring the confidentiality of party identities. To enhance its applicability, the authors propose two future research directions: integrating PrivIdEx with Hyperledger Fabric's identity-mixer-based certificate authority to prevent collusion attacks, and generalizing PrivIdEx for other confidential assets.

6. In Reference 51 the authors proposed a biometric identification scheme based on zk-SNARKs that prevented biometric template information leakage. The proposed scheme was a solution to schemes that stored biometrics templates directly, which were insecure and schemes that applied feature transformations or biometric encryption, which showed reduced identification accuracy. By using zk-SNARKs the proposed scheme could create proofs that collected fingerprint features matched a known biometric template without leaking said biometric template. Blockchain technology was additionally used to provide transparency on access policy evaluations and to prevent tampering. The security analysis included in the research described the security aspects of the system and showed the correctness of the proposed solution regarding the security aspects. The performance analysis of the proposed scheme demonstrated that the threshold size and number of fingerprint features impact accuracy and false rejection rate. The scheme's execution times increased linearly with the number of fingerprints, while accuracy remained constant.

The efficiency and privacy protection aspects of the proposed scheme were seen as its main benefits. However, to improve the usefulness of the scheme the authors mentioned some future research directions including designing protocols based on the proposed scheme, further improving the scheme's efficiency, and designing application scenarios.

7. In Reference 52 the authors conceptualized accountable attribute-based authentication with fine-grained access control (AccABA). Subsequently, they proposed an attribute-based, fair, anonymous, and publicly traceable crowdsourcing scheme on the blockchain using AccABA. With the introduction of AccABA the authors solved the privacy problems posed by regular authentication schemes, as well as the accountability and trusted third parties problems in anonymous authentication schemes. To ensure that a ciphertext could be decrypted only for users matching the access policy, AccABA used Ciphertext-policy attribute-based encryption (CP-ABE). Then using zk-SNARKs a match with the access policy could be proven without disclosure of the user attribute values.

A security analysis was included which described security properties that the system had to conform to, as well as proofs on how these security properties were fulfilled. The security analysis included a comparison of the properties (e.g., anonymous, fair, accountable) of the proposed system with several other crowdsourcing systems.^{53–57} Aside from the security analysis, the authors did not include other analyses including the performance analysis. Such analysis was likely not included because the researchers did not implement an experimental system implementation to conduct the performance analysis. The authors mentioned the combination of privacy, accountability, and fairness as the main benefits of the system.

8. In Reference 58 the authors proposed LaT-Voting, an anonymous and traceable decentralized voting scheme that is based on the blockchain. Traditional e-voting systems were dependent on a centralized platform which introduced the risk of tampering. Blockchain-based solutions contrarily aimed to solve this problem but were in turn unable to detect double-voting. The introduced notion of “prefix-based linkable-and-traceable anonymous authentication” used in LaT-Voting aimed to solve both the tampering and double-voting, while the scheme preserved user privacy and did not require a trusted third-party. The proposed solution uses zk-SNARK zero-knowledge proofs and smart contracts in blockchain technology to implement functionality. Smart contracts collect cast ballots, verify their validity, and detect double-voters. Blockchain addresses serve as pseudonyms for user identities, enabling vote immutability and public verification of the voting process. Voters register with a certificate authority, obtain a certificate with keys, and generate an authentication token using a zk-SNARK. The generated tokens are verified for validity, and if two tokens have a common prefix, they are considered double-votes. The public key is derived from the secret key obtained from combining two different authentications from a single voter.

The research included a security analysis in which the authors described the correctness of the proposed solution. The authors also explained how the scheme obtained the desired privacy, anonymity, unforgeability, public verifiability, and traceability features. They further proved the correctness and security of the scheme. For the security, they did so by first describing a theorem for unforgeability, anonymity, linkability, and accountability, after which they proved the four theorems and concluded the security of the proposed scheme. In conclusion, the main benefit of the LaT-Voting scheme was that it enabled decentralized and anonymous voting, while it was able to detect double-votes and trace them back to a malicious voter.

9. In Reference 59 the authors introduced a privacy-protecting authorization system that leveraged blockchain technology and the zk-SNARK protocol. By ensuring that sensitive attribute data remained undisclosed the proposal addressed a critical concern with existing authorization solutions for blockchain resources. The system employed zk-SNARKs to generate authorization proofs based on identity attributes, ensuring that attribute values were not

revealed. Ethereum smart contracts then verified the correspondence between the blockchain address used during proof creation and the proof user, thereby limiting authorization usage to the proof generator.

The authors conducted a cost analysis that demonstrated the relatively higher expense of their proposed system compared to alternative solutions such as UPort⁶⁰ and EverSSDI.⁶¹ Additionally, a security analysis was included which outlined potential attacks on the proposed solution including tampering with credentials and smart contracts. However, the authors proved these attacks to be computationally or practically infeasible. While the primary advantage of the proposal highlighted by the authors was privacy protection, its major drawback was the substantial operational cost. To mitigate this expense, the authors suggested future research directions involving the utilization of alliance chains instead of the main Ethereum blockchain.

10. In Reference 62 the authors propose BIMP, a blockchain-based incentive mechanism for location proofs, to address privacy concerns in location-based services. Current systems, such as GPS or centralized parties, can cheat, leak location information, and be vulnerable to attacks. BIMP aims to generate privacy-preserving location proofs, include an incentive mechanism for witnesses, and detect collusion attacks, making nearby nodes more willing to participate. BIMP implemented functionality using hash-based commitments and ring signatures to generate location proofs (LPs) between a witness and a prover. Blockchain technology addressed scalability, rewards, and synchronization issues. Zero-knowledge proofs, specifically zk-SNARKs, were used by provers to prove ownership without revealing their identity. The traceable-detectable-prefix structure prevents prover-witness collusion attacks by using simple hashes to detect attacks while hiding the identities of the prover and witness.

The authors conducted a security analysis on BIMP, demonstrating its ability to resist attacks, achieve unforgeability, privacy, proof of ownership, and unlinkability, and its effectiveness in collusion detection. Also included was a comparison with two alternative schemes, STAMP⁶³ and PROPS.⁶⁴ For this comparison, the authors used a “co-efficient of satisfaction (CS)” formula based on five features: “privacy protection”, “collusion prevention”, “safety employing”, “data backup”, and “incentive mechanism”. The results of this comparison showed that independent of the number of LPs, BIMP reached a higher “CS” than both STAMP and PROPS because of its greater level of privacy and security.

The authors concluded with some benefits of BIMP, namely that it effectively incentivized witnesses to generate LPs and effectively protected user privacy. In the end, the authors suggested an investigation into the storage problem of LPs in the generation phase as a future research direction, however, the authors did not seem to elaborate on this suggestion.

11. In Reference 65 the authors proposed pRate, a reputation management scheme that provided strong privacy and security guarantees. Reputation systems in online platforms were an important measure of trustworthiness. However, in traditional reputation schemes, there were no integrity guarantees, and neither guarantee was user privacy. On top of that some users did not want to leave a bad rating in fear of receiving a low rating in retaliation. While previous schemes attempted to solve these problems, none of them also hid the reputation from the reputation manager. To that end, pRate was proposed to solve the problems with previous schemes while also hiding the reputation from the reputation manager. The reputation manager could however still learn the user identities of transacting parties such that users could report misbehaving parties. The pRate scheme utilizes technologies like BBS+ signatures, Chaum-Pedersen-signed ElGamal (CPS-EG) encryption, and zero-knowledge proofs, including bulletproofs, to create user reputation credentials. These credentials can be used to disclose messages or create zero-knowledge proofs. The scheme also uses bulletproof zero-knowledge proofs to prove interval-based statements on user reputation. CPS-EG is used to ensure the secrecy of rating tokens and ratings. The research conducted a theoretical performance analysis of a scheme, providing formulas for calculating proof size and computational duration of various protocol operations. The exact values were inferred by comparing the parameters of the formula with those in the system. Additionally, The research conducted a security analysis on pRate, identifying five security theorems, which were proven to be true.

The main benefit of pRate was that it comprised a reputation system that hid identities from participants in the transaction and rating phases, and additionally did not allow the reputation manager to learn the reputation scores either. While the reputation manager knew the identities of transaction participants, this benefited the scheme because it enabled users to report misbehaving participants. Finally, the authors mentioned that the pRate scheme functioned independently from the actual transactions, and as such pRate could be used with other approaches for anonymous transaction and payment schemes.

12. In Reference 66 the authors proposed an attribute-based privacy-preserving identity management system on the blockchain. This identity management system revealed real identities upon consensus of a system policy violation,

which solved the need to always disclose identity attributes to central identity providers. Other identity management systems require this identity attribute disclosure for verification and traceability purposes. The proposal used blockchain technology, zk-SNARKs, ElGamal encryption, and hash-based commitments to achieve selective attribute disclosure, anonymity, and unforgeability. The same technologies also prevented collusion attacks, while the system's traceability requirement was instead satisfied using Shamir's secret sharing together with blockchain technology.

The authors conducted a security analysis, proving unforgeability, anonymity, and traceability. They also conducted performance and cost analyses using an off-chain simulation of their proposed system. The performance analysis assessed the system's time and space complexity, while the cost analysis compared it to Ethereum's basic transactions, finding that transactions were 1.24 to 189.42 times more expensive in the proposed system. The authors highlight the benefits of anonymity and traceability in their proposed system but also acknowledge its limitations, such as high computational power, potential collusion attacks, and reduced system functionality. They suggest future research on efficiency improvements and malicious off-chain validator detection techniques.

13. In Reference 67 the authors propose a blockchain-based federated learning system that prioritizes fair incentives, integrity, and privacy. This system enhances existing federated learning systems by preventing data leakage, adding plausibility checks on contributed models, and preventing data-poisoning and free-riding attacks. The proposal utilized Local Differential Privacy (LDP), zk-SNARKs, and blockchain technologies to create plausible deniability, verify model updates on real private data, and distribute the federated learning system, ensuring neutrality, transaction immutability, and transparency. These technologies prevent leaks of private data through pattern information.

The performance analysis showed that key generation times, proof generation times, and circuit size scale linearly with the number of constraints, while verification duration, verification key size, and proof size remain constant. Although a security analysis was not included, the authors acknowledged that attacks on integrity could occur if the learning task was known beforehand. The authors analyzed the cost of on-chain operations on the Ethereum blockchain, stating it was prohibitively expensive, and suggested using permissioned blockchains like Quorum for lower costs and higher throughput.

The main benefits of the proposed solution were its confidentiality, fairness, and tamper-resistance, while the solution was also scalable and decentralized. In the end, the authors mentioned some future research directions including improving the performance and scalability by using zk-STARKs and a recursive verification system, as well as researching alternative incentive mechanisms, and ensuring system applicability in practice.

14. In Reference 68 the authors proposed a privacy-preserving biometric authentication scheme for authenticating to a server without a trusted third party. To this end, they proposed and compared three proof of decryption techniques based on HMAC, blinding techniques, and verifiable computing (VC). These techniques were an alternative to the use of trusted execution engines (TEEs) to obtain the required trust for biometric authentication. The proposed scheme used Pinocchio, a practical zk-SNARK, in the verifiable computing technique to verify a secret key without exposing it. A conducted performance analysis compared the proposed VC scheme with the matrix-vector multiplication and determinant-based blinding alternatives. Memory consumption of the VC scheme was shown to be relatively low. On the contrary, execution times were orders of magnitude higher than those of the other two techniques.

The authors additionally mentioned the security level of the scheme in bits, together with the brute-force success probability. However, no comprehensive security analysis was performed. The elimination of the TEE or trusted third party was the main benefit of the proposed method. Suggested future research directions included the implementation of HMAC-based schemes, as well as testing alternative VC techniques for use in the scheme.

15. In Reference 69 the authors proposed the DECO (DECentralized Oracle) system. DECO allowed a user to prove statements, such as that a piece of data came from a particular website, about data accessed from a TLS-protected website, without revealing the data itself. By enabling a user to prove such statements, the user could securely access their data, export it, and then prove the data origin to another service. By doing so DECO removes the need for help from the origin service and solves limitations of existing solutions with similar ideas that require deprecated TLS versions or server-side installation to function. The functionality of DECO was enabled by using zk-SNARKs and a three-party handshake protocol. The zk-SNARKs were used to prove the statements on retrieved data without disclosing the data to the verifier, while the three-party handshake protocol let the prover and verifier from a joint TLS client. This joint TLS client provided authenticity and prevented the prover from secretly encrypting additional

messages using the TLS session keys. Security analysis was provided that included simulation-based proofs of the stated security theorem.

Additionally, a performance analysis was performed using a demo implementation of three different applications. The analysis showed the run duration of the different protocols on LAN and WAN, offline and online, and showed the time and memory costs of generating and verifying the zero-knowledge proof steps. At the end of the performance analysis, the authors compared the end-to-end performance of DECO to Town Crier,⁷⁰ which was approximately twenty times faster than DECO but required more trust assumptions. The compatibility of the proposed system was mentioned as its main benefit. While DECO showed reduced performance compared to Town Crier, the ever-so-often usage of DECO made the authors mention that this trade-off was acceptable for the security gain. Though the authors did not provide specific future research directions, they did mention that DECO could with some adaptations use Bulletproofs to remove the trusted setup requirement. However, using Bulletproofs would have come at the cost of increased proof creation and verification times.

The most prevalent solutions in this category describe (attribute-based) authentication and identity applications with References 47, 50, 51, 52, 59, 66, and 68. This prevalence appears logical given that these we specifically searched for authentication and identity systems in this systematic literature review. Furthermore these solutions can be used for multiple types of categories which makes that they are classified in this category. Still, we can observe several differences between the works and how they aim to achieve authentication. For example, References 51 and 68 specifically focus on biometric authentication and identification, where in Reference 51 the ZKPs is used to directly verify the fingerprint against the biometric template while keeping it private while Reference 68 uses ZKPs to verify knowledge of a secret key in verifiable computing without exposing this key. In general, the most used task of ZKPs in the proposed attribute-based identity schemes is to verify the identity and attributes match what is expected, without revealing identity details.

In References 48 and 65 we see a similar topic of reputation management systems. These solutions provide privacy by using ZKPs to keep identities hidden, while preventing users the traditional flaw of users dishonestly resetting their reputation through adoption of a new, clean identity.

Another group of works with somewhat related goals focused on agreements, specifically on Reference 49 focusing on consensus protocols⁴⁹ or voting systems.⁵⁸ In the consensus protocol the importance of ZKPs lies in allowing participation in proof of stake without revealing information on a user's monetary assets, while in the voting system the ZKPs ensure that votes remain private while guaranteeing that each voter is only allowed to vote exactly once, otherwise invalidating duplicate votes.

Works in References 62 and 67 both included some intention to use ZKPs for fair incentives. This can be for location-based services⁶² where a user is remunerated for proving proximity in which the ZKP guarantees the exact location remains private, or for providing useful training data in federated learning,⁶⁷ where the ZKP keeps the contents of the training data private while guaranteeing certain statements on this data are valid. This latter solution relates to 27 in the medical category which proposed the medical data-market for machine learning solutions in the medical category, which also kept the medical data private.

As for author suggestions on where to go next the recommendations were all over the place, if the authors indicated any further directions in the first place. This makes sense since the applications in this category are categorized for their general use, which means the author may think about some specific situations that other that other authors did not. They might also intent to further generalize the solution for even broader use.

3.9 | Other

We review different use cases in this section that do not fall under any of the mentioned categories:

1. In Reference 71 the authors proposed an efficient, privacy-preserving data-sharing framework for vehicles using multi-sharding blockchain. This solution was proposed to solve performance, security, and privacy concerns in vehicular data sharing schemes that included: a single point of failure, identity disclosure, and requirement for a trusted third party. The main enabling technologies in the proposal were zk-SNARK and blockchain technology. zk-SNARKs were used to enable auditable data sharing without disclosure of vehicle identities. To enable said data sharing when scaling to many vehicles, the proposal used blockchain technology for a multi-sharding blockchain protocol that increased

performance and scalability. The included security analysis first described three attack types, and then the authors proved that the proposed system was secure against the described attacks.

Additionally, the proposal included a performance analysis on theoretical (big-O notation) and practical performance. The practical performance was evaluated for both sharding and multi-sharding proof-of-concept implementations and showed the bandwidth consumption and throughput for different numbers of blockchain nodes in the system. The practical performance analysis also showed run durations of the proposed scheme in comparison to ring and group signature-based schemes. The main benefits of the proposed framework are its security, privacy, and performance aspects. The authors concluded by suggesting research on the block sorting mechanism as a future research direction.

2. In Reference 72 the authors proposed a privacy-preserving fair payment (PPFP) protocol for use in vehicle-to-grid (V2G) networks. V2G, where electric vehicles exchange electricity with a smart grid, raises privacy concerns due to the transmission of private information. Existing solutions have limitations like high computational complexity and communication overhead. PPFP, using blockchain technologies and zero-knowledge proofs, aims to address these limitations, providing lower computational complexity and stronger privacy without relying on a trusted third party. PPFP uses nested commitment schemes to hide transaction amounts and participants, unlike Bitcoin-based timed commitments (BBTCs). PPFP prevents identity leaks using zk-SNARKs. The authors found PPFP to satisfy privacy, fair payment, and untraceability properties, describing two theorems on privacy and untraceability and fairness, which were proven to hold.

The main benefits of PPFP were that it allowed EVs and the smart grid to exchange services and electricity without relying on trusted hardware or a central authority. PPFP also satisfied the three properties of privacy, fairness, and untraceability. In contrast to these benefits was the downside of the time-consuming Bitcoin transactions, however the authors diminished this downside by mentioning that PPFP could use alternative consortium blockchains without the loss of security.

3. In Reference 73 the authors proposed Eunomia, a blockchain-based vehicular digital forensics (VDF) scheme that was secure and privacy-preserving. Gathering evidence for VDF was of great importance for investigations into vehicular accidents and crimes. It was therefore a major problem when witnesses withheld evidence for fear of retaliation, data leaks, and data misuse. By setting requirements on privacy (anonymity and unlinkability) and security (confidentiality, authentication, fine-grained access control, accountability, and traceability) for Eunomia, the authors aimed to resolve privacy concerns in witnesses and thereby improve the effectiveness of VDF. The proposal in Eunomia aimed to meet privacy and security requirements using technologies like zero-knowledge proofs, Pedersen commitments, ciphertext-policy attribute-based encryption (CP-ABE), and blockchain technologies. Bulletproof was used for warrant validity, while smart contracts validated these proofs. CP-ABE ensured only encrypted data requests were acknowledged. Pedersen commitments were used in an oblivious transfer protocol to embed the investigator's private key in data, providing traceability and embedding the investigator's private key.

A security analysis in the research showed that the anonymity, unlinkability, confidentiality, and authentication requirements of the scheme were met. To prove them, the authors described a theorem for each of the requirements and subsequently proved this theorem. For the fine-grained access control, accountability, and traceability requirements the authors instead provided an explanation of how they were achieved. Furthermore, the authors included a comparison of the privacy and security properties of the proposed solution compared to previous works, including DiaLOG,⁷⁴ DFAV,⁷⁵ B4F,⁷⁶ and BB-VDF.⁷⁷ Only Eunomia was able to achieve all the compared security and privacy properties in this comparison. The authors conducted a security analysis and performance evaluation of Eunomia, a prototype on a Wi-Fi-based Ethereum test network. They found unfavorable results, possibly due to improved privacy and security. They also included a cost analysis on Ethereum gas costs.

To conclude, the authors mentioned that the main benefit of Eunomia was that data providers could participate in VDF without privacy concerns. At the same time investigation tasks were not disclosed, data could not easily be altered, and data leaks could be traced. A downside of Eunomia was that the crowdsourcing of data enabled false information to be uploaded to the system. However, the authors described that the effects of this downside were mitigated because data providers would be asked to testify in court, as would have been the case for witnesses in justice systems throughout the world.

4. In Reference 78 the authors proposed ZXAD, an abuse detection system for private Tor exit nodes that did not reveal any information apart from a high traffic volume. The problem with Tor was that while it could be used for legitimate purposes, it was also abused for high-traffic attacks including denial of service (DoS) attacks. This led to

services blocking Tor exit node IPs, which in turn hindered the legitimate use of Tor from that exit node. Existing solutions were either challenging to implement for Tor because of its unlinkability trait or posed a privacy risk to Tor users. The proposed solution instead implemented a privacy-preserving connection identification system which allowed the detection of high-volume attacks. The detected attacks could then be blocked, which would mean that services could unblock Tor access for legitimate users without the fear of high-volume attacks from the same origin. Zero-knowledge proofs, specifically zk-SNARKs, were the main enabling technology in ZXAD. A client used these proofs to prove their unique ID to obtain a long-term key, without revealing the ID itself. The obtained key was then used to generate periodic keys for generating the circuit and stream tokens. Zero-knowledge proofs were then generated on these tokens again to prove that they were well formed. The tokens allowed to rate limit the number of connections in a period while preserving the user privacy. BLS signatures additionally guaranteed the validity of key distribution messages. Furthermore, Shamir's secret sharing was used to extend ZXAD for Tor's t-out-of-n threshold threat model. This threat model guaranteed the security of ZXAD if a majority of the Tor DirAuths were honest.

The research conducted a security analysis to verify the accuracy of zero-knowledge proofs used to verify circuit and stream tokens. It also evaluated the performance of the ZXAD protocol using Kyber and Libsnark libraries, focusing on execution and verification times, blind signature transfer, circuit token generation phases, and load for DirAuths, exits, and clients. The authors identified several limitations in the ZXAD protocol, including the need for appropriate unlinkable connections, stream tokens sent to destination servers, destruction of a common reference string initial secret, and potential DoS attacks by clients. However, the main advantage of ZXAD was its ability to limit high-volume traffic attacks while maintaining strong privacy guarantees.

5. In Reference 79 the authors proposed V2GEx, a privacy-preserving fair vehicle-to-grid (V2G) exchange scheme based on blockchain technology. By reducing the privacy-preservation aspect to only one party the authors then also proposed Uni-V2GEx, an alternative scheme that offered better efficiency than regular V2GEx. V2G solutions allow electric vehicles (EVs) to improve grid stability by exchanging electricity and serving as grid regulators. A critical issue for this service however was the required constant monitoring of EVs for service regulation and rewards that led to the identity and location information of EV owners being compromised. The two proposed schemes solved this issue by using blockchain technology combined with zero-knowledge proofs to preserve privacy while achieving fair exchange in V2G. The two proposals utilized blockchain technology, specifically smart contracts, as a distributed broker for micro-payments, replacing third-party intermediaries. Zk-SNARKs were used to verify transaction accuracy and maintain privacy. Uni-V2GEx, unlike V2GEx, only used zk-SNARKs for the EV side, reducing privacy for the smart grid. This improved efficiency by eliminating zk-SNARK proof computations for the smart grid side. The authors provided a formal security proof for V2GE, a privacy analysis that explained how it thwarted traffic analysis, de-anonymization attacks, and attacks on payment, identity, and location privacy. They also implemented a prototype on Ethereum, conducted a performance analysis of zk-SNARKs, micropayments, and transaction processing, and added a cost analysis of the prototype.

To conclude, the authors mentioned that the main benefit of V2GEx was that it simultaneously provided fairness and bi-directional privacy, while Uni-V2G provided improved efficiency compared to V2GEx for situations where unidirectional privacy was adequate.

6. In Reference 80 the authors proposed a hybrid identity authentication scheme for blockchain-based mobile crowd sensing (MCS). By including a hierarchical authentication model to perform computation off-chain and verification on-chain, the proposed solution increased the privacy and efficiency of the system. In contrast, existing blockchain-based systems exposed node identities leading to reduced privacy, or had low verification efficiency and failed to scale with the blockchain. The proposal used zk-SNARKs to perform proof-generating computations for authentications off-chain, which were then verified on-chain in an efficient and privacy-preserving manner. Additionally, blockchain technology was used to decentralize the MCS system while smart contracts on the blockchain authenticated nodes within a cluster. An included security analysis of the proposed system started with a threat model with potential attacks and risks for the system and then described how the system prevented these threats. Afterward, the security of the proposed solution was favorably compared against other schemes.^{81–84}

Additionally, a performance analysis was included, which listed the performance of registration, witness generation, proof generation, and proof verification. The proof verification performance was then favorably compared against that of an electric vehicle verification scheme.⁸² Next to the performance analysis, a cost analysis showed the costs

of performing the operations for registering nodes on the Ethereum blockchain. The authors remarked on the privacy and security aspects of the system as its main benefits and suggested mechanisms for managing MCS participant reputation and methods for sharing collected data as future research directions.

Given the category designation of Other and the relatively low number of works included in this category there is not much overlap between the directions of the works category. Therefore we will keep this paragraph relatively short.

The two works that are most related both proposed a solution for providing remunerations in vehicle-to-grid.^{72,79} In such V2G network the providing vehicle needs to be constantly monitored for the energy they provide to and request from the network to manage the load on the electricity grid. The difference between these two works lies in Reference 72 focusing on hiding the monetary value of transactions and enforcing a fair payout, while Reference 79 furthermore concentrates on the privacy aspect of the vehicle identity and location and additionally describes two versions of the proposal. One where privacy is preserved on both the vehicle and grid side, as well as a less computationally demanding version for when privacy on the side of the electrical grid is not required.

While References 71 and 73 are also proposals that relate to the use of ZKP in an application involving vehicles, they are strictly different from the V2G proposals, and each other, since Reference 71 is a solution for information exchange between vehicles in a manner that retains vehicle identity privacy while Reference 73 is a solution for digital forensics to research accidents and crimes involving vehicles. The solution in the latter work uses ZKPs as one of the techniques to retain anonymity and unlinkability while simultaneously enforcing access control, accountability and traceability of investigators and their data access.

Finally, Reference 80 is a work that somewhat relates to the previously described works on authentication and identity in the General category. However, it was instead classified in this category as the work strictly focuses on identity authentication in mobile crowd sensing, and is therefore not directly applicable to other use cases.

Two works provide recommendations on future actions,^{71,80} both of which focus on further enhancing the proposed solutions with feature improvements.

4 | RESULTS

In this section, we delve into the outcomes of the literature review. The Subsection 4.1 discusses the results derived from the evaluation of the research works included, as conducted in Section 3.4. Subsequently, we present a performance assessment in Section 4.2, and a security evaluation for those that carried out a security analysis in Section 4.3, respectively. In the end, we revisit the research questions proposed in Section 2.2, incorporating all our findings and providing answers in Section 4.4.

4.1 | General findings

Out of the 41 research papers reviewed, 31 employed zk-SNARKs, and 10 made use of Bulletproofs. Notably, none of the research included in our systematic literature review utilized zk-STARKs. Although we cannot draw overarching conclusions about the comparative usage of zk-STARKs versus zk-SNARKs and Bulletproofs, it is evident that zk-STARKs are not commonly employed in privacy-preserving applications research that adheres to our search and filtering criteria. The Bulletproof protocol's security relies on the discrete logarithm problem, which makes it vulnerable to quantum attacks, as Shor's algorithm⁸⁵ has demonstrated the ability to solve discrete logarithm problems efficiently using quantum gates.⁸⁶ Similarly, homomorphic public-key cryptography is not quantum-resistant, which means that the zk-SNARK protocol also lacks quantum resistance. In our systematic literature review, all the analyzed works utilize one of these two protocols, leading us to the conclusion that none of these solutions can be considered post-quantum secure without additional protective measures. Surprisingly, with the exception of two research works, none of the studies included in our review made any mention of the quantum resistance of their proposed solutions. Furthermore, these mentions were brief and lacked in-depth exploration. Although a few research papers acknowledged their dependency on the discrete logarithm problem for security in their security analysis, none of the authors related this to quantum resistance. Only research papers, References 18 and 39 briefly acknowledged and considered the quantum resistance of their proposed solutions, but even these works did not delve into the potential impact on solution security or provide insights into how to approach quantum resistance.

We will continue our findings with a comparison of the real-world performance and security of the proposed solutions.

4.2 | Performance comparison

In this section, we explore the performance aspects of the NIZKP protocol implementations in the reviewed research papers and conduct a comparative analysis of performance metrics, provided by the authors in their performance assessments. Our goal is to reach conclusions regarding performance disparities among the different protocols. The performance analysis results are listed in Table 8.

The table uses the following abbreviations for column names to make it fit on a single page:

1. **Paper**—The literature source in the corresponding use case grouping
 - a. **Fin.** Financial
 - b. **Med.** Medical
 - c. **Bus.** Business
 - d. **Oth.** Other
 - e. **Gen.** General
2. **Pr.**—Protocol; The protocol used in the research
 - a. **B** Bulletproof (rows coloured grey)
 - b. **SN ZK-SNARK**
3. **Type** Defines how the performance metrics were obtained. Includes system component, parameter configuration, and test machine specifications
4. **ST (ms)** Setup Time; The time required to set up the system in milliseconds
5. **KT (ms)** Keygen Time; The time required for generating the keys in milliseconds
6. **PKS (B)** Proving Key Size; The storage size of the proving key in bytes
7. **VKS (B)** Verification Key Size; The storage size of the verification key in bytes
8. **PS (B)** Proof Size; The storage size of the generated proof in bytes
9. **Txs (B)** Transaction Size; The size of a generated transaction in bytes
10. **PT (ms)** Proof Time; The time required to generate a proof in milliseconds
11. **VT (ms)** Verification Time; The time required to verify a proof in milliseconds

To keep this concise, we did not include every performance result from each work. Instead, we selected the most pertinent results, including multiple types when they were substantially different (e.g., different system components). When the work examined a single component with multiple parameters, we chose the parameters with detailed descriptions in the text, or we estimated the results from graphs if necessary. In cases where parameters overlapped among different system components, we incorporated the performance values for both the lowest and highest parameters tested for each component. The research works furthermore conducted performance analyses on differing hardware. In Table 7, we list for each work the specifications of the system the researchers used to conduct the performance analysis as indicated by the authors of the work. One work used two different machines, for each of those machines we defined a type in Table 7 which we also referenced in the type column of Table 8.

Financial use case: Out of the six research works within the financial use case category, five conducted a performance analysis. Among these, four employed the zk-SNARK protocol, while only one utilized Bulletproofs. It's important to note that the performance analysis in Reference 23 is excluded from our comparison because it assessed the computational overhead of blockchain versus a traditional database, rather than comparing zero-knowledge proof protocols. Consequently, we had four performance analyses in total for comparison: three based on the zk-SNARK protocol and one using the Bulletproof protocol. As can be seen in Table 8, the proof size of the bulletproof in Reference 18 is more than twice as large as the proof size in Reference 22. Similarly, the bulletproof transaction size is somewhere from slightly larger to more than three times larger than the transaction size generated by the zk-SNARKs in Reference 24. The proof generation times of the bulletproof in Reference 18 are anywhere from just under three times as fast compared to the fastest zk-SNARKs results in Reference 24, to over 20 000 times as fast compared to the slowest zk-SNARKs results from Reference 25. The opposite is true for proof verification times, though the differences are much smaller. For this metric, the bulletproof proof results from Reference 18 were anywhere from slightly slower compared to the zk-SNARK results in Reference 22, to just over 15 times slower compared to the fastest zk-SNARK result in Reference 24. From this small

TABLE 7 Machine specification.

Paper	Type	System specifications
Financial		
18	i7 i5	i7-4870HQ, 16GB RAM
22		i7-8750, 12GB RAM
		i5-4200H, 12GB RAM
24		i7-7700, 16GB RAM
25		AWS t2.large, 8GB RAM
Medical		
28		i7-4770M, 8GB RAM
30		64x E5-4610 v2, 128GB RAM
32		i7-8700k, 32GB RAM
34		i5-11400H, 16GB RAM
Business		
39		M1, 16GB RAM
43		unspecified “common computer”
44		Xeon Silver 4210, 16GB RAM
45		i7-9750H, 8GB RAM
Other		
78		i7-6700k (single thread), ? RAM
79		i7-8700, 16GB RAM
80		i5-1135G7, 8GB RAM
General		
47		i7-8750H, 16GB RAM
48		i7-3632QM, 16GB RAM
50		i7-4790, 16GB RAM
51		unspecified i5, 8GB RAM
66		i5-4210M, 4GB RAM
67		AWS 16vCPU, 64GB RAM
69		AWS c5.2large 8vCPU, 16GB RAM

sample size, we can inconclusively say that bulletproofs generate larger proof and transaction sizes than zk-SNARKs. The same can be seen for the verification times of bulletproofs, where bulletproof proofs are slower to verify than proofs generated using zk-SNARKs. In contrast, bulletproofs are up to several orders of magnitude faster in generating the proofs than zk-SNARKs.

Medical use case: Of the seven research works included in the medical use case category, six included a performance analysis. Four of these six works used zk-SNARK as the protocol while two used bulletproofs. However, the performance analysis in Reference 26 is not usable in our comparison since it analyses the performance of deploying smart contracts instead of the zero-knowledge proofs. Similarly, Reference 31 only performed an analysis on the response times of the system and did not provide performance numbers for the zero-knowledge proofs which we could compare here. This means that in total we had four performance analyses to compare, three using the zk-SNARK protocol and one using the Bulletproof protocol. As listed in Table 8, there were only two aspects on which we can compare the bulletproof in Reference 30 to the zk-SNARK in the other research works, namely on proof generation time and proof verification time. Both times show that bulletproofs are somewhere in the middle regarding speed compared to the zk-SNARKs in the other works in the category. The proof generation time of the bulletproof is more than 18 times

TABLE 8 Performance analysis results.

Paper Fin.	Pr.	Type	ST (ms)	KT (ms)	PKS (B)	VKS (B)	PS (B)	Txs (B)	PT (ms)	VT (ms)
18	B	Confidential transaction						1310.00	40.00	14.00
	B	Auditing policies limit policy					622.00		21.50	7.50
22	SN	zkChain, i7		5931.78	12193219.20	573.25	286.75		1789.68	5.21
	SN	zkChain, i5		8966.48	12193219.20	573.25	286.75		2788.52	6.77
24	SN	BN256, Base Tx	730.00					418.00	110.00	0.89
	SN	BN256, Tx both limits	3400.00					802.00	430.00	0.92
	SN	BLS12-381, Base Tx	1300.00					546.00	180.00	1.50
	SN	BLS12-381, Tx both limits	6200.00					1122.00	730.00	1.60
25	SN	Commitment			6140000.00	511.80			10000.00	
	SN	First claim, MerkleTreeDepth = 16			103660000.00	511.80			131000.00	
	SN	First claim, MerkleTreeDepth = 64			447480000.00	511.80			466000.00	
Med.										
28	SN	1000 circuit inputs		16300.00	15300000.00	31500.00	125400.00		52000.00	614.00
30	B	FES-SPH-PSM Bulletproofs							262.20	168.00
32	SN	1000 users		3148.00					2018.00	219.00
	SN	20 000 users		99026.00					18257.00	181.00
34	SN	FAD phase 10 inputs							1757.00	133.10
	SN	FAD phase 60 inputs							4920.00	988.10
	SN	PPT phase, 2 ¹⁰ message space							41.10	22.20
	SN	PPT phase, 2 ⁶⁰ message space							83.20	40.50
	SN	IPP phase 16-bit ID							14.00	10.00
	SN	IPP phase 512-bit ID							810.00	611.00

(Continues)

TABLE 8 (Continued)

Paper Fin.	Paper Pr.	Type	ST (ms)	KT (ms)	PKS (B)	VKS (B)	PS (B)	TxS (B)	PT (ms)	VT (ms)
Bus.										
39	B	Inclusion proofs, 1 aggregated range					672.00		8.00	2.00
	B	Inclusion proofs, 256 aggregated ranges					1184.00		2005.00	381.00
43	SN	128-bit security, 1M-gate-circuit, 1000-bit input		117 000 to 123 000			288.00		147 000 to 784 000	< 10
44	B	Standard range 1-p-1-v mode, 8-bit data size					1088.00		28.00	14.00
	B	Standard range 1-p-1-v mode, 64-bit data size					1568.00		197.00	91.00
	B	Arbitrary range 1-p-1-v mode, 8-bit data size					1312.00		52.00	25.00
	B	Arbitrary range 1-p-1-v mode, 64-bit data size					1792.00		388.00	175.00
	B	Bidding, 128-bit bidlength							642.11	4.74
45	B	Aggregation and winner verification, 512 bidders							3480.00	11.18
Oth.										
78	SN	Libsnark implementation zk-SNARK proof					169.00		3270 ± 20	8.4 ± 0.1
79	SN	Convert	18 000		63000000.00	633.00			6000	30 000
	SN	Commit	23 000		85000000.00	596.00			7000	24 000
	SN	Claim/Refund	11 000		32000000.00	559.00			3000	31 000
	SN	Deposit	11 000		11 5000000.00	670.00			3000	31 000
	SN	Pour(Zerocash)	31 000		898000000.00	749.00			8000	22 000
80	SN	100 circuit inputs							5000	10840.00
	SN	600 circuit inputs							9000	10840.00
Gen.										
47	SN	Modular approach			11 000 000		700		3000	
48	SN	Mint statement		34,40	3230.00	820.00	287.00		25.20	22.40
	SN	Update statement		42600.00	49240000.00	1849.00	287.00		22100.00	21.00
50	SN	Circuit 2, 64-bit nonce		12865.90	32903000.00	3285.60	280.00		3380.00	4.50
	SN	Circuit 2, 128-bit nonce		12872.90	32938000.00	3597.00	280.00		3382.70	4.50

(Continues)

TABLE 8 (Continued)

Paper Fin.	Paper	Pr.	Type	ST (ms)	KT (ms)	PKS (B)	VKS (B)	PS (B)	TxS (B)	PT (ms)	VT (ms)
51		SN	9 fingerprints	26.79						46.92	1.82
		SN	54 fingerprints	186.95						318.77	23.78
66		SN	Generate PKc, VKc		106240.00	192005879.00	6497.00				
		SN	Generate PKd, VKd		2830.00	3896140.00	3381.00				
		SN	Generate π_C					566.00		36620.00	
		SN	Generate π_D					566.00		1580.00	
67		SN	$\pi_W, k = 4, n = 100$		177000.00		25000.00	708.00		8641.00	236.00
		SN	$\pi_W, k = 4, n = 1000$		1360000.00		25000.00	705.00		74945.00	232.00
		SN	$\pi_K, k = 4, n = 100$		174000.00		4200.00	705.00		8152.00	244.00
		SN	$\pi_K, k = 4, n = 1000$		1380000.00		4200.00	705.00		73696.00	243.00
69		SN	Binary option phase					861.00		12970 $\pm$ 40	10.00
		SN	Age proof phase					574.00		3670 $\pm$ 20	10.00
		SN	Price discrimination phase					1722.00		12680 $\pm$ 20	50.00

slower than the fastest result in Reference 34, but equally more than 198 times faster than the result in Reference 28. The proof verification times show comparable results, where the bulletproof is more than 16 times slower and five times faster than the fastest and slowest zk-SNARK results respectively, both from Reference 34. The results from the small sample size in this category show mixed results, with the bulletproof protocol being faster than the zk-SNARK protocol for some application types while being slower for others. These results can have many reasons, which we will discuss in Section 5. In short, though, we suspect the difference in protocol application to be the main differentiating factor.

Business use case: The business use case category included eight research works, of which only four included a performance analysis. From the results in Table 8 we can see that a single research work used zk-SNARKs as opposed to the bulletproofs used by other works. The proof size generated by bulletproof in Reference 43 was less than half the size of the smallest zk-SNARK proof in the category. The proof generation time of the zk-SNARK proof however was at least 42 times slower than that of the slowest bulletproof in Reference 45. Finally, the proof verification times of the only zk-SNARK in this category were listed as below 10ms, so we cannot compare this value exactly. The value however is comparable to the fastest bulletproof proof generation time of 2 ms from Reference 39, while up to about two orders of magnitude slower than the slowest bulletproof proofs in References 39 and 44. The results from the small sample size in this category show that the zk-SNARK has a smaller proof size and about equal or lower verification times than the bulletproofs. The zk-SNARK proof generation time however was much slower than the Bulletproof proof generation times from other works in this category.

General use case (multiple use cases): In the general use case category, there were 14 research works of which 10 included a performance analysis. Of the works that included such performance analysis, nine used zk-SNARK as the NIZKP protocol while only one used bulletproofs. However, the performance analysis in Reference 62 focuses on the performance of the collision detection, not on the performance of the zero-knowledge proof protocol as we want to compare here. The authors of Reference 65 did not concentrate on the NIZKP protocol's performance either, providing only theoretical performance formulas rather than actual numerical values. As a result, we were unable to compare these two research works' performance analyses. Additionally, we are unable to properly evaluate⁶⁸ because it appears to cover more ground than only ZKP performance and primarily concentrates on completely homomorphic encryption in the performance analysis. As such, we were unable to appropriately compare the performance analysis's findings. This indicates that the zk-SNARK technique was used in all seven performance studies that we could compare. As a result, we are unable to compare the variations among the protocols in this use case group. However, we can use the same approach to examine the differences between works, with wildly differing outcomes. For the key generation time, proof generation times, and verification key size, we see differences of around 40 000, 3000, and 30 times between the lowest values from Reference 48 and highest values from Reference 67 respectively. Furthermore, the largest proving key size from Reference 66 is nearly 60 000 times larger than the smallest proving key size from Reference 48. Finally, the longest proof verification time from Reference 67 is about 134 times larger than the shortest time from Reference 51, while the smallest proof size from Reference 50 is about six times smaller than the largest proof size from Reference 69. In short, the verification key sizes, proof sizes, and even the verification times show a reasonable deviation in our opinion, while the other values show a variance that is much higher than we would expect.

Other use case: The six research works in the other use case category included five works with performance analysis. Of the works that included such performance analysis, four used the zk-SNARK NIZKP protocol while only one used the Bulletproof protocol. However, the performance analysis in Reference 71 focuses mostly on the blockchain. While a running time comparison that showed the running time of various parts of the system was included, these were the numbers for the general scheme and not specifically the zero-knowledge proof protocol. Similarly, Reference 73 included a thorough performance analysis that focused on the computation performance and communication overhead of the entire system, the results of which mostly speak to complete actions in the system which include too much interaction with Ethereum for us to compare the performance. This means that in total we had three performance analyses to compare, all using the zk-SNARK protocol. Therefore, we could not compare the differences between protocols within this use case category. However, we investigated the difference between works using the same protocol, and we compared two criteria, that is, proof generation and verification times. These numbers show that the proof generation times between References 80, 78, and 79 are within the same order of magnitude. The same cannot be said about the proof verification times, where the times from Reference 78 are about three orders of magnitude lower than,⁷⁹ while the numbers in Reference 79 were already up to five times slower than the proof verification times in Reference 80.

Overview: In our systematic literature review, we examined 41 research works, with 30 of them including a performance analysis. Among these 30 works, 22 utilized the zk-SNARK NIZKP protocol, while the remaining eight works employed the bulletproof protocol. We were able to compare the performance of zero-knowledge proof protocols in 22 research works across different use case categories. Out of these, five research works analyzed the performance of the bulletproof protocol, while the other 17 works focused on the zk-SNARK protocol. The examination of zero-knowledge proof (ZKP) protocols across various use cases and implementations uncovers a rich tapestry of performance metrics, revealing nuanced trends and trade-offs. Firstly, the setup time, which denotes the duration required to initialize the system for generating proofs, exhibits notable disparities among different protocols. For instance, the bulletproof protocol demonstrates an impressive minimum setup time of 730 milliseconds (ms), underscoring its efficiency in this aspect. Conversely, certain implementations, such as those utilizing zkChain, may entail longer setup times, which can extend to several seconds.

Moving beyond setup time, the key generation phase represents another crucial aspect of ZKP protocols. Key generation time, often influenced by factors like cryptographic parameters and hardware specifications, varies significantly across different scenarios. This underscores the importance of optimizing for the used hardware in enhancing protocol efficiency.

Moreover, the size of generated proofs is a critical consideration, particularly in resource-constrained environments. Here, the analysis reveals diverse outcomes, with proof sizes ranging from relatively modest to more substantial dimensions. Protocol choices and specific use case requirements heavily influence proof size, with some implementations favoring smaller proofs for improved scalability and reduced storage overhead.

In addition to proof size, the time required to generate and verify proofs constitutes another pivotal performance metric. Across the spectrum of analyzed protocols, proof generation and verification times exhibit variability, reflecting trade-offs between computational complexity and security assurances. Notably, certain implementations achieve remarkably swift proof generation times, such as the zkChain protocol on an i7 processor, where proof verification times were a mere 5.21 ms. However, verification times may vary, with some protocols necessitating longer durations for cryptographic validation.

These findings underscore the intricate interplay between performance metrics and protocol design choices in the realm of zero-knowledge proofs. While certain implementations excel in specific aspects like proof generation efficiency, others may prioritize different parameters, such as proof size or verification time. Ultimately, the selection of an optimal ZKP protocol hinges on a nuanced understanding of application requirements, encompassing factors like security, scalability, and computational efficiency. Engineers can harness the power of zero-knowledge proofs to achieve robust privacy guarantees without compromising performance by sensibly navigating these considerations.

4.3 | Security comparison

This section examines security aspects in 41 research works, comparing results to determine differences in protocols, 32 included security analysis, categorized into financial, medical, business, general, and other categories. Works are grouped based on analysis type, with some performing multiple types. The first grouping we define includes works that provided security definitions in theorems or lemmas and then proved these. In this grouping, we define three sub-groupings based on the type of the provided proofs. The first subgroup proved the theorems or lemmas by describing a proof in natural language and includes References 50 and 65. The second subgroup proved the theorems or lemmas by (additionally) providing a more mathematical proof. This includes proofs where a mathematical description was an important part of the proof, or where the authors provided a formal mathematical proof. This included References 18, 34, 36, 45, 48, 69, 72, 73, and 79. The final subgroup proved the theorems or lemmas not just through mathematical proofs, but specifically through what are called security games. This group includes References 24, 32, and 66.

The second group we designate consists of publications that list certain security specifications or security features of the offered solution, followed by an explanation of how these specifications or system features were satisfied. We may classify the majority of the works in this category into a single subgroup where the descriptions of how the requirements or aspects were satisfied were primarily written in everyday language. References 22, 28, 31, 42, 43, 58, 52, 62, 73, and 79 are members of this subgroup. A second grouping used mathematical formulas to explain how the security standards or aspects were partially or completely met. The members of this subgroup are References 25, 30, and 45.

A third group we define includes publications that listed a number of attacks on the proposed solution, and subsequently described the security measures in place to prevent these attacks or how these attacks could otherwise be

mitigated. We define two subgroups in this group as well. The first provided just general descriptions of the attacks and the security measures using mostly natural language. This includes References 22, 43, 47, 71, and 80. The other subgroup consists of publications that give these descriptions partly or fully in a mathematical way. This subgroup includes References 59 and 78.

A fourth group we define includes works that separately defined a threat model in the security analysis of their work. This group includes References 36 and 44.

In conclusion, two publications offered a security analysis that other papers did not. One group provided an overview of the system's security. Only the work of Guo et al.⁵¹ is included in that group, wherein they used mathematics to characterize the soundness and completeness of the zk-SNARK and demonstrate the security of their suggested fix. An additional group computed the likelihood of the suggested solution's security failing. Ji and Chalkias³⁹ computed the failure probability of their suggested solution by mathematical means in the single study that is included in this group. Despite the fact that we detailed multiple methods by which the security analyses of the included works were carried out, few works carried out the security analysis in a comparable manner. Upon reflection, it became evident that comparing these security analyses posed a challenge due to inherent differences in methodologies. However, rather than viewing this as a limitation, we recognize it as a possible direction for future research and standardization efforts. One actionable question that arises from our analysis is: *How can future security analyses of non-interactive zero-knowledge proof (NIZKP) application implementations be standardized to facilitate better comparison?* Addressing this question could lead to the development of standardized frameworks or methodologies for evaluating security in NIZKP implementations, thus enabling more meaningful comparisons across different studies as we shall address in more detail in Section 5

4.4 | Answering the research questions

Our systematic literature research attempts to address the research issues we outlined in Section 2.2. In this section, we restate the questions and provide an answer.

1. What are the existing real-world use case implementations of zk-SNARKs, zk-STARKs, and bulletproofs in privacy-preserving authentication, as documented in the literature? How are these protocols currently being applied in practical scenarios, and what insights can be drawn from these implementations regarding their effectiveness, challenges faced during deployment, and potential improvements for broader adoption?

In this systematic literature review work, we included 41 research works that implement at least one of the zk-SNARK, zk-STARK, or bulletproof protocols in a privacy-preserving solution. We found these works using the search query defined in Subsection 2.3.1 and narrowed them down to include only those that seek to protect privacy. The search query that needed either “authentication” or “identity,” even if our filtering did not contain any filtering rules on the authentication components, ensured that the application was at least connected to authentication. In Section 3.4 we analyze each work and provide a use case description. In Table 6 we also provide the column Application which is further summarized in a few words for which the solution using NIZKPs is used.

1. What are the comparative performance and security implications of implementing zk-SNARKs, zk-STARKs, and bulletproofs in privacy-preserving authentication systems? This includes an analysis of authentication speed, computational overhead, and resilience against various types of attacks, as well as potential vulnerabilities.

The real-world performance of the NIZKP protocol implementations in the included works and suggested solutions were compared in Subsection 4.2. Once more, since we were unable to include any works that employed zk-STARKs in our SLR, we are unable to discuss their performance in real life here. As previously mentioned, although there was a considerable to extremely big fluctuation in the outcomes, we discovered that the real-world performance of the bulletproof protocol and zk-SNARK protocol was comparable. With the exception of one extreme outlier, the most trustworthy finding indicates that proofs produced by the bulletproof protocol were often larger than those produced by the zk-SNARK protocol. The result that the bulletproof protocol generates a proof faster than the zk-SNARK protocol, however, is not so clear. Since no works that employed zk-STARKs were included in our SLR, we are unable to discuss their performance in real life here. As previously mentioned, although there was a considerable to

extremely big fluctuation in the outcomes, we discovered that the real-world performance of the bulletproof protocol and zk-SNARK protocol was comparable. With the exception of one extreme outlier, the most trustworthy finding indicates that proofs produced by the bulletproof protocol were often larger than those produced by the zk-SNARK protocol.

Verification time results are the least dependable because they vary the most. Overall, it appears that both protocols require roughly the same amount of time to validate a proof; but, depending on application specific factors, either of the two protocols may show to be faster than the other when comparing the protocols in different applications. To put it briefly, we are unable to come to a definitive conclusion about which protocol performs better in actual situations. In Section 5, we go into further detail on the significance of these findings and our next moves. Furthermore, we were unable to offer any real-world results on the implementations' security. The security studies in the included publications vary too much from one another, as described in Subsection 4.3, to allow any conclusions about the security differences between the bulletproof and zk-SNARK protocols to be made. The one security conclusion we could draw—that neither zk-SNARKs nor bulletproofs are quantum resistant—is covered in further detail in Section 5. This implies that if quantum security is a concern, zk-STARKs are still the sole choice available among the three protocols. The choice of zk-SNARK, zk-STARK, or bulletproof protocol in an application involves two main trade-offs: performance, security, and applicability. The theoretical performance of these protocols is vastly different, but real-world performance does not show clear differences. Security differences are not clear due to mutual differences between security analyses. The zk-SNARK and bulletproof protocols are not quantum resistant, introducing an implicit vulnerability to quantum computer attacks. The zk-STARK and bulletproof protocols furthermore do not require a trusted setup, which is another security aspect to consider. The applicability trade-off is more important for applications involving value ranges. The bulletproof protocol is a range protocol, making it more desirable for applications involving value ranges. Other trade-offs include the availability of libraries that implement each NIZKP protocol, but these are not discussed in this work.

5 | DISCUSSION

In this section, we discuss the results of our systematic literature review (SLR). We first, investigate the strengths and limitations of our work in Subsection 5.1. We provide an overview of the aspects our SLR did well and where other researchers could use our work. Then to contrast these strengths we mention some of the limitations that we ran into with our work, where our work is not that useful for researchers, and where other research can expand further on our work. Furthermore, we show that our SLR work provides a novel contribution in Subsection 5.2 through a comparison with related SLR works. Finally, Subsection 5.3 details some potential future research directions that we either observed from the included works or that we formed ourselves based on the results and conclusions of our work.

5.1 | Strengths and limitations

Our SLR as performed in this work has several strengths and weaknesses. In this section, we list what we conceive to be the main strengths and limitations of our work.

5.1.1 | Strengths

1. In this work, we provided, through means of a SLR, an elaborate overview of current research into applications that use non-interactive zero-knowledge proofs (NIZKP) to preserve privacy. We conceive this to be the main strength of our work since it provides a good overview of the different applications of the main NIZKP protocols for privacy-preserving applications. Such an overview is beneficial to see the big picture of the possibilities enabled by NIZKPs.
2. More importantly, this work provides an overview of applications such that researchers can see immediately whether a solution to their problem using NIZKPs already exists, which allows these researchers to build upon and improve previous works.

3. Our work is also beneficial in case any researchers want to perform research on NIZKPs but do not know where to start. In that case, the overview of the applications using NIZKPs for their devised solution can give researchers innovative ideas.
4. More specifically, the future research directions given by most works allow researchers to immediately find a concrete research topic and goal.
5. Our work is furthermore valuable in providing an overall picture of the status of non-interactive zero knowledge proofs, including how often each of the three included NIZKP protocols is used and for which general applications. For the same applications, our work also provides the typical performance to expect from the solution using ZKPs if the original research provided these figures.
6. Finally, our SLR gives an idea of the status of research on using NIZKPs for privacy-preserving authentication applications. For example, we noted that comparisons between the different applications and NIZKP protocols are difficult because there currently does not seem to be a standardized way to conduct performance analyses. And while for the performance analysis at least the type of the recorded metrics correspond, for example, proving key size, proof generation time, and so forth, the security analyses do not show such correspondence. The way the authors analyzed the security of the application, and which security aspects they analysed, are vastly different between different research works. This makes it hard for researchers to compare the security of the different research works, even if the applications themselves are comparable.

5.1.2 | Limitations

Aside from its strengths, our work also has several limitations. The primary constraint lies in our search methodology, where despite comprehensive efforts, the search query, filter rules, and inclusion criteria failed to yield any works utilizing the zk-STARK protocol. This omission significantly impacts the SLR's ability to compare three NIZKP protocols comprehensively, hindering our ability to draw conclusive results. Despite conducting an additional literature search on October 10, 2023, with refined parameters excluding zk-SNARK and Bulletproof terms, we encountered the same limitation, with no new works found. Tables A1 and A2 in the Appendix A details the reasons for not including the six newly found works. The diversity of applications across the included works presents another limitation, making it challenging to compare the use of NIZKPs effectively. The varying technical approaches and solutions adopted by researchers for different applications hinder the generalizability of our findings. Additionally, discrepancies in security analyses, ranging from differing assessment methodologies to variations in the components analyzed, further complicate comparisons between works. Despite these limitations, our SLR provides valuable insights into the diverse ways in which NIZKPs are applied and security analyses are conducted in the field. However, the lack of uniformity in the computational environments used for performance evaluations also poses a challenge, as differences in machine specifications may skew the results, rendering direct comparisons between studies difficult.

5.2 | Related systematic literature review research

In this section, we compare our SLR with the works of other authors addressing related topics involving various types of reviews. Through this comparison, we discuss the differences in our approach, thereby highlighting the unique contribution of our SLR. To start off we compare our SLR with one of the works rejected for inclusion in this work since it did not propose an application that used NIZKPs.⁸⁷ This work instead compared the zk-SNARK and zk-STARK protocols including an overview of implementing libraries. The authors then described how zk-SNARKs could be used for identity management on the blockchain, provided examples of an identity attribute compliance use case, and evaluated some performance measurements on the examples. In short, we can conclude that while the work also aimed at the privacy-preserving authentication/identity topic, there are several differences with our SLR. First, this work only compared the zk-SNARK and zk-STARK protocols and did not include the bulletproof protocol. On top of this, the authors only compared the protocols on the technical aspects and libraries that implement them, not on the performance. Finally, our work includes many different applications in the privacy-preserving authentication/identity space, while this work only includes an individual use case of an identity system on the blockchain.

As described, our SLR differentiates itself from this work through the inclusion of three ZKP protocols and comparing these on the performance and security for multiple privacy-preserving authentication/identity use cases. During our research, we only found a single work that performed some kind of systematic literature review of zero-knowledge proofs (ZKPs), we sought and included more similar systematic literature review works using the following two search queries:

“zero-knowledge” AND “literature review”

and

“zero-knowledge” AND “survey”

Subsequently, we examined outcomes from works closely aligned with our work for meaningful comparisons. It's important to highlight that not all works are compared in this section; rather, we selectively incorporated the most pertinent findings. To the best of our knowledge, the systematic literature reviews presented in this section are the most akin to our work in terms of comparability. Numerous located works, such as Reference 88 and one by Li et al.,⁸⁹ delved into ZKPs comprehensively, providing historical context and detailed mathematical insights. These works proceeded to examine the potential applications of ZKPs. Chen et al.'s⁹⁰ work centered on IoT applications, while Gowravaram's⁹¹ focused on ZKP applications in financial regulation. The key distinction between these works and ours lies in our specific focus on three NIZKP protocols: zk-SNARKs, zk-STARKs, and bulletproofs. Our emphasis then shifted to comparing the real-world performance disparities of these protocols in applications with a privacy-preserving authentication use case. Although works like Kurmi et al.'s⁹² and a work by Pathak et al.⁹³ shared a similar setup by focusing on authentication use cases, they diverged from our approach by not comparing the three specific protocols and their respective performance metrics.

Within Cerulli's work,⁹⁴ a literature review explored the mechanics of ZKPs and group signatures. Subsequently, the study proceeded to formulate ZKPs tailored for various applications. In contrast, our SLR distinguished itself by evaluating three specific ZKP protocols, analyzing their real-world application scenarios, and assessing their implementation performance based on insights derived from other works. Subsequently, two research works diverged from ZKP reviews: Herbowo's,⁹⁵ which delved into self-sovereign identity systems, and Herskind et al.'s,⁹⁶ which explored privacy-preserving approaches for electronic cryptocurrencies. While these works incorporated ZKPs, unlike our SLR, these proofs were not the primary focus. Consequently, the authors did not engage in a comparative analysis of the three specific NIZKP protocols concerning real-world performance.

Two of the reviewed works we encountered focused on a singular NIZKP protocol. Specifically, Chen et al.'s⁹⁷ work centered on zk-SNARK, while Morais et al.'s⁹⁸ delved into bulletproofs. These reviews delved more extensively into the intricacies of their respective protocols compared to our work, providing insights into potential applications. However, in contrast to our approach, these works did not undertake a comparative analysis of multiple protocols regarding their real-world performance in applications. Additionally, the applications discussed in both works diverged from our emphasis on privacy-preserving authentication. There were also studies that, like our SLR, compared multiple NIZKP protocols. For instance, Sun et al.'s⁹⁹ work delved into both the zk-SNARK and Bulletproof protocols, providing insights into their theoretical performance using big-O notation. However, unlike our work, they did not delve into real-world performance. The applications discussed in their study were exclusively related to blockchain use, whereas our focus on privacy-preserving authentication extended beyond the blockchain. Similarly, Partala et al.¹⁰⁰ conducted a comparison of various NIZKP protocols, including zk-STARKs and bulletproofs, assessing both theoretical and practical performance. However, their performance comparison focused directly on the protocols rather than evaluating the real-world performance of solutions, as our SLR does. Additionally, their emphasis was on applications in blockchain, in contrast to our broader scope.

In another study, Panait et al.⁸⁷ compared zk-SNARK and zk-STARK protocols but omitted the comparison of the Bulletproof protocol, which is a distinctive feature of our work. Their investigation centered on libraries implementing these protocols, a facet not covered in our study. Conversely, our work extensively compares the application use cases for ZKP protocols and evaluates the real-world performance of these solutions, aspects not explored in their work.

Finally, two studies most closely aligned with ours were an analysis by Gong et al.¹⁰¹ and an investigation incorporated in a work by Sanchez Ortiz.¹⁰² Gong et al.'s analysis meticulously examined the zk-SNARK, zk-STARK, and bulletproof protocols, providing an in-depth exploration of their mathematical underpinnings and big-O performance. Furthermore, the analysis delved into the quantum resistance of the protocols. Nonetheless, it did not compile or contrast applications for these NIZKP protocols. Additionally, their study omitted a comparison of real-world

performance—a key objective of our SLR. In the literature review within Sanchez Ortiz's work, there was an exploration of the general workings and cryptographic fundamentals of ZKPs. It also drew comparisons between the zk-SNARK and bulletproof protocols, particularly in finance use cases, with a discussion on the zk-STARKs protocol as well. The primary distinction from our study is that our focus is on privacy-preserving authentication, whereas their emphasis is on financial applications. Moreover, our research goes a step further by comparing solutions utilizing Non-Interactive ZKPs (NIZKPs) in other works based on real-world performance. In contrast, their approach involves using existing literature to explore the application of ZKPs for guidance in developing their own use case implementations. We intended to incorporate a comparison with Khandekar's work,¹⁰³ however, due to the unavailability of the full text, we were unable to access it and, consequently, could not include their work in this comparison.

Based on the aforementioned comparison with other works, we infer that our systematic literature review stands out as distinct from previous studies, to the best of our knowledge. Consequently, we assert that our work represents a novel contribution to the research on the application of non-interactive ZKPs, particularly within the domain of privacy-preserving authentication.

5.3 | Future research directions

While not all the works included in our analysis offered insights into future research directions, among those that did, one of the most frequently mentioned focuses was the optimization of proposed solutions for enhanced efficiency and speed. Notably, a common aspect identified as a source of slow performance was ZKPs, particularly in scenarios where faster alternatives using traditional cryptographic methods could achieve similar functionality. Some works that were excluded during the systematic literature review filtering process did not leverage ZKPs and instead opted for more conventional cryptographic methods for performance reasons. The authors of works choosing ZKPs defended this decision by highlighting the additional privacy-preserving aspects that ZKPs bring to their proposed solutions. The observed lower performance and efficiency of ZKPs emerged as a deliberate trade-off for the enhanced privacy-preserving aspects they offer in a solution. Consequently, delving into ways to enhance the performance and efficiency of non-interactive NIZKPs emerges as a promising avenue for future research. This exploration has the potential to improve the performance of NIZKPs, rendering the trade-off more acceptable and thereby increasing the attractiveness of choosing NIZKPs for the enhanced privacy they can provide. Noteworthy works advocating for this future research direction include References 21, 28, 32, 47, 51, 66, and 67.

Another avenue for future research, as proposed by References 26, 39, and 50, involves exploring the applicability of the proposed solution in different domains or for alternative applications. These authors advocated for such investigations, believing that a solution employing ZKPs could offer significant privacy benefits, potentially extending its usefulness beyond the specific subject explored in their respective works. Moreover, nearly all the reviewed works suggested, or at least implied, that subsequent research on their solutions would be beneficial, especially when limitations were identified. This may include implementing alternative options for specific components of the system or modifying the solution to mitigate potential attacks that were discovered.

In addition to the future research directions outlined in the reviewed works, our systematic literature review led us to identify another potential avenue. In the realm of quantum-resistant cryptography, also known as post-quantum cryptography,¹⁰⁴ significant developments have occurred. The American National Institute of Standards and Technology (NIST) initiated a call for proposals in 2016 to standardize post-quantum cryptographic algorithms, ultimately revealing four quantum-resistant cryptographic algorithms in 2022 for general encryption or digital signature applications.¹⁰⁵

Considering these advancements, we propose that researching the quantum resistance of NIZKP protocols is a prudent undertaking. This entails understanding how quantum resistance alters the dynamics of NIZKPs and its implications for applications, as described in the works included in our systematic literature review. One such example of a quantum-resistant NIZKP protocol is the zk-STARK protocol.¹⁰⁶ However, it is noteworthy that despite its quantum resistance, the zk-STARK protocol is not widely utilized, as none of the works in our review incorporated it into their solutions. Given this observation, we put forward that exploring the integration of quantum-resistant NIZKP protocols as a replacement for non-quantum-resistant protocols, aimed at achieving post-quantum security, represents a valuable future research direction.

In addition to the current applications of ZKPs discussed in this paper, such as privacy-preserving authentication, financial transactions, and medical data sharing, recent advancements have highlighted the potential of ZKPs in the field of deep learning.^{107–109} Specifically, ZKPs have garnered attention for their ability to verify the correctness of computations in deep learning models without compromising data privacy. Future research directions in this area could explore various applications of ZKPs in deep learning, including model verification, privacy-preserving inference, outsourced computations, model aggregation, and data ownership verifications. By leveraging the unique properties of ZKPs, such as privacy preservation and verifiability, researchers can develop innovative solutions to address privacy concerns and ensure the integrity of deep learning models and data. Furthermore, investigating the scalability and efficiency of ZKP-based approaches in the context of large-scale deep learning applications could open new avenues for enhancing data privacy and security in machine learning systems. Overall, the integration of ZKPs into deep learning holds promise for advancing the field towards more trustworthy and privacy-preserving AI solutions, and further exploration of this area represents a compelling direction for future research.

Finally, we observed the lack of a benchmarking and security comparison standard for ZKPs in Section 4.3, and hypothesize already that such standardized procedures would provide benefits to researchers in looking to compare research works on ZKPs. We therefore believe research into the possibility to define aforementioned standardized benchmarking and security comparison procedure for ZKP applications, including the benefits, downsides, and intricacies of such standard, to be a promising future direction.

5.4 | Conclusion

This study analyzed the wide field of privacy authentication, focusing on the application of zk-SNARK, zk-STARK, and Bulletproof NIZKP protocols. Through a detailed systematic literature review, we examined 41 research works that leverage these protocols across diverse applications. Our objective is twofold: to evaluate the real-world performance and test the security implications of these implementations. In dissecting the performance metrics, we uncover a wide spectrum of outcomes, highlighting the variability in efficiency across different implementations. However, our attempts to compare the security analyses encounter challenges resulting from the diverse methodologies employed by researchers. Despite these challenges, we provide a comprehensive discussion that explains the strengths and limitations of our approach, positioning our work as a distinctive contribution to the existing literature on zero-knowledge proofs. Looking ahead, we highlighted the future research directions drawn from both the reviewed studies and our insights. Our proposed plan includes creating a strong real-world benchmark designed specifically for NIZKP protocols. This benchmarking project aims to tackle unanswered questions left by this systematic literature review, paving the way for more progress in privacy authentication research.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Mohammed El-Hajj  <https://orcid.org/0000-0002-4022-9999>

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APPENDIX

TABLE A1 Excluded research literature.

Title	DOI	Reason
A Privacy-Preserving Payment Model for EV Charging	10.1007/978-3-030-89432-0_21	No protocol usage
AMVchain: authority management mechanism on blockchain-based voting systems	10.1007/s12083-021-01100-x	No protocol usage
An Efficient Group Signature Based Digital Currency System	10.1007/978-981-15-2767-8_34	No protocol usage
An efficient identity tracing scheme for blockchain-based systems	10.1016/j.ins.2021.01.081	No protocol usage
An organization-friendly blockchain system	10.1016/j.cose.2019.101598	No protocol usage
Anonymous Traceability protocol based on Group Signature for Blockchain	10.1016/j.future.2021.09.020	No protocol usage
Blockchain based privacy-preserving software updates with proof-of-delivery for Internet of Things	10.1016/j.jpdc.2019.06.001	No protocol usage
Blockchain-Based Self-Sovereign Identity System with Attribute-Based Issuance	10.1007/978-3-031-21280-2_2	No protocol usage
Casper: a blockchain-based system for efficient and secure customer credential verification	10.1007/s42786-021-00036-3	No protocol usage
Computing Neural Networks with Homomorphic Encryption and Verifiable Computing	10.1007/978-3-030-61638-0_17	No protocol usage
Decentralized Access Control Encryption in Public Blockchain	10.1007/978-981-15-2777-7_20	No protocol usage
Decentralized Privacy-Preserving Netting Protocol on Blockchain for Payment Systems	10.1007/978-3-030-51280-4_9	No protocol usage
DECOUPLES: a decentralized, unlinkable and privacy-preserving traceability system for the supply chain	10.1145/3297280.3297318	No protocol usage
DEPLEST: A blockchain-based privacy-preserving distributed database toward user behaviors in social networks	10.1016/j.ins.2019.05.092	No protocol usage
Dynamic pricing in industrial internet of things: Blockchain application for energy management in smart cities	10.1016/j.jisa.2020.102615	No protocol usage
Efficient Novel Privacy Preserving PoS Protocol Proof-of-concept with Algorand	10.1145/3475992.3475999	No protocol usage
EIFFeL: Ensuring Integrity for Federated Learning	10.1145/3548606.3560611	No protocol usage
ElearnChain: A privacy-preserving consortium blockchain system for e-learning educational records	10.1016/j.jisa.2021.103013	No protocol usage
emmy-Trust-Enhancing Authentication Library	10.1007/978-3-030-33716-2_11	No protocol usage
FAST: Fair Auctions via Secret Transactions	10.1007/978-3-031-09234-3_36	No protocol usage
hOCBS: A privacy-preserving blockchain framework for healthcare data leveraging an on-chain and off-chain system design	10.1016/j.ipm.2021.102535	No protocol usage

(Continues)

TABLE A1 (Continued)

Title	DOI	Reason
HyperNet: A conditional k-anonymous and censorship resistant decentralized hypermedia architecture	10.1016/j.eswa.2022.118079	No protocol usage
Kicking-the-Bucket: Fast Privacy-Preserving Trading Using Buckets	10.1007/978-3-031-18283-9_2	No protocol usage
KPH: A Novel Blockchain Privacy Preserving Scheme Based on Paillier and FO Commitment	10.1007/978-981-19-5209-8_7	No protocol usage
Lattice-Based Zero-Knowledge Proofs: New Techniques for Shorter and Faster Constructions and Applications	10.1007/978-3-030-26948-7_5	No protocol usage
Lockmix: a secure and privacy-preserving mix service for Bitcoin anonymity	10.1007/s10207-019-00459-6	No protocol usage
MiniLedger: Compact-Sized Anonymous and Auditable Distributed Payments	10.1007/978-3-030-88418-5_20	No protocol usage
NECTAR: non-interactive smart contract protocol using blockchain technology	10.1145/3194113.3194116	No protocol usage
Nummatus: A Privacy Preserving Proof of Reserves Protocol for Quisquis	10.1007/978-3-030-35423-7_10	No protocol usage
Oblivious Message Retrieval	10.1007/978-3-031-15802-5_26	No protocol usage
PACHain: Private, authenticated & auditable consortium blockchain and its implementation	10.1016/j.future.2020.05.011	No protocol usage
PACHain: Private, Authenticated and Auditable Consortium Blockchain	10.1007/978-3-030-31578-8_12	No protocol usage
PASTRAMI: Privacy-preserving, Auditable, Scalable & Trustworthy Auctions for Multiple Items	10.1145/3423211.3425669	No protocol usage
PPCA: privacy-preserving conditional actions for IoT environments using smart contracts	10.1145/3360774.3360794	No protocol usage
Privacy and information sharing in a judicial setting: a wicked problem	10.1145/2757401.2757425	No protocol usage
Privacy Preserving Biometric Authentication on the blockchain for smart healthcare	10.1016/j.pmcj.2022.101683	No protocol usage
Privacy-Enhancing Decentralized Anonymous Credential in Smart Grids	10.1016/j.csi.2020.103505	No protocol usage
Privacy-preserving energy storage sharing with blockchain and secure multi-party computation	10.1145/3508467.3508471	No protocol usage
Privacy-Preserving Incentive Systems with Highly Efficient Point-Collection	10.1145/3320269.3384769	No protocol usage
Private and Trustworthy Distributed Lending Model Using Hyperledger Besu	10.1007/s42979-021-00500-3	No protocol usage
QHSE: An efficient privacy-preserving scheme for blockchain-based transactions	10.1016/j.future.2020.06.025	No protocol usage
Quantifying location privacy in permissioned blockchain-based internet of things (IoT)	10.1145/3360774.3360800	No protocol usage

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TABLE A1 (Continued)

Title	DOI	Reason
Random-Value Payment Tokens for On-Chain Privacy-Preserving Payments	10.1007/978-3-031-17834-4_13	No protocol usage
SecTEP: Enabling secure tender evaluation with sealed prices and quality evaluation in procurement bidding systems over blockchain	10.1016/j.cose.2021.102188	No protocol usage
Simple and scalable blockchain with privacy	10.1016/j.jisa.2020.102700	No protocol usage
Sunspot: A Decentralized Framework Enabling Privacy for Authorizable Data Sharing on Transparent Public Blockchains	10.1007/978-3-030-95384-3_43	No protocol usage
Towards a privacy-preserving smart contract-based data aggregation and quality-driven incentive mechanism for mobile crowdsensing	10.1016/j.jnca.2022.103483	No protocol usage
TPPSUPPLY: A traceable and privacy-preserving blockchain system architecture for the supply chain	10.1016/j.jisa.2022.103116	No protocol usage
VERICONDOR: End-to-End Verifiable Condorcet Voting without Tallying Authorities	10.1145/3488932.3497758	No protocol usage
Verifiable Computation in Multiparty Protocols with Honest Majority	10.1007/978-3-319-12475-9_11	No protocol usage
Vulnerability market as a public-good auction with privacy preservation	10.1016/j.cose.2020.101807	No protocol usage
ZeroLender: Trustless Peer-to-Peer Bitcoin Lending Platform	10.1145/3374664.3375735	No protocol usage
zkSk: A Library for Composable Zero-Knowledge Proofs	10.1145/3338498.3358653	No protocol usage
A Group Signature Based Digital Currency System	10.1007/978-981-15-2777-7_1	Indirect protocol usage
A Privacy-Preserving Framework for Endorsement Process in Hyperledger Fabric	10.1016/j.cose.2022.102637	Indirect protocol usage
An E-Voting System Based on Tornado Cash	10.1007/978-3-031-25467-3_8	Indirect protocol usage
Succinct Attribute-Based Signatures for Bounded-Size Circuits by Combining Algebraic and Arithmetic Proofs	10.1007/978-3-031-14791-3_31	Indirect protocol usage
Zero-Knowledge Proofs for Set Membership: Efficient, Succinct, Modular	10.1007/978-3-662-64322-8_19	Indirect protocol usage
RingCT 3.0 for Blockchain Confidential Transaction: Shorter Size and Stronger Security	10.1007/978-3-030-51280-4_25	Optional protocol usage
Zero-Knowledge for Homomorphic Key-Value Commitments with Applications to Privacy-Preserving Ledgers	10.1007/978-3-031-14791-3_33	Optional protocol usage
Accountable Privacy for Decentralized Anonymous Payments	10.1007/978-3-662-54970-4_5	Indirect protocol usage, adapts zk-SNARKs for their application
An access control model for the Internet of Things based on zero-knowledge token and blockchain	10.1186/s13638-021-01986-4	Privacy preservation not a (direct) main goal, describes improving IoT access control using Blockchain

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TABLE A1 (Continued)

Title	DOI	Reason
Blockchain-Based Confidential Payment System with Controllable Regulation	10.1007/978-3-031-21280-2_3	No detailed usage, only evaluation mentions Bulletproofs
Circuitree: A Datalog Reasoner in Zero-Knowledge	10.1109/ACCESS.2022.3153366	Application to more easily write zk-proofs, not use them
DBS: Blockchain-Based Privacy-Preserving RBAC in IoT	10.1007/978-3-030-91424-0_6	No detailed usage
Designing a blockchain-enabled privacy-preserving energy theft detection system for smart grid neighborhood area network	10.1016/j.ejpsr.2022.107884	No detailed usage
Efficient Verifiable Image Redacting based on zk-SNARKs	10.1145/3433210.3453110	Indirect protocol usage, uses CP- and CC-SNARK
Efficient Zero-Knowledge Proofs on Signed Data with Applications to Verifiable Computation on Data Streams	10.1145/3548606.3560630	Indirect protocol usage, uses CP-SNARK
Non-Interactive Zero-Knowledge Proofs for Composite Statements	10.1007/978-3-319-96878-0_22	Indirect protocol usage, proposes a new protocol
Offline Authentication Scheme based on Blockchain Technology for Smart Lock	10.1145/3291842.3291893	No detailed usage, just mentions zk-SNARK can be used
On Using zk-SNARKs and zk-STARKs in Blockchain-Based Identity Management	10.1007/978-3-030-69255-1_9	No proposed application, compares zk-SNARK and zk-STARK
Privacy-Preserving Analytics for Data Markets Using MPC	10.1007/978-3-030-72465-8_13	No detailed usage
Provisions: Privacy-preserving Proofs of Solvency for Bitcoin Exchanges	10.1145/2810103.2813674	No protocol usage, uses sigma-protocols. Full version uses zk-SNARK, but no detailed usage
Public Verifiable Private Decision Tree Prediction	10.1007/978-3-030-71852-7_16	Indirect protocol usage, proposes a new NIZKP
RZee: Cryptographic and statistical model for adversary detection and filtration to preserve blockchain privacy	10.1016/j.jksuci.2022.07.007	Indirect protocol usage, proposes improvements on zk-SNARK
Zapper: Smart Contracts with Data and Identity Privacy	10.1145/3548606.3560622	Indirect protocol usage, uses SE-SNARK
ZeroCross: A sidechain-based privacy-preserving Cross-chain solution for Monero	10.1016/j.jpdc.2022.07.008	Indirect protocol usage, uses CP-SNARK
ZeroMT: Multi-transfer Protocol for Enabling Privacy in Off-Chain Payments	10.1007/978-3-030-99587-4_52	No detailed usage
Zether: Towards Privacy in a Smart Contract World	10.1007/978-3-030-51280-4_23	Indirect protocol usage, proposes new protocol based on sigma-protocols and Bulletproofs
Zk-AuthFeed: How to feed authenticated data into smart contract with zero knowledge	10.1109/Blockchain.2019.00020	Indirect protocol usage, uses zk-DASNARK
zkCNN: Zero Knowledge Proofs for Convolutional Neural Network Predictions and Accuracy	10.1145/3460120.3485379	Indirect protocol usage, uses CP-SNARK

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TABLE A1 (Continued)

Title	DOI	Reason
ZkrpChain: Privacy-preserving data auditing for consortium blockchains based on zero-knowledge range proofs	10.1109/TrustCom50675.2020.00092	Less complete version of included full paper “zkrpChain: Towards multi-party privacy-preserving data auditing for consortium blockchains based on zero-knowledge range proofs”
ZPiE: Zero-Knowledge Proofs in Embedded Systems	10.3390/math9202569	Privacy preservation not a (direct) main goal, shows usage in low-power IoT

TABLE A2 Extra zk-STARK literature search; found and excluded research literature.

Title	DOI	Reason
A regulated anonymous cryptocurrency with batch linkability	10.1016/j.csi.2023.103770	No protocol usage for a privacy-preserving application
A Review of Privacy-Preserving Cryptographic Techniques Used in Blockchain Platforms	10.1007/978-3-031-29857-8_23	No protocol usage
BV-ICVs: A privacy-preserving and verifiable federated learning framework for V2X environments using blockchain and zkSNARKs	10.1016/j.jksuci.2023.03.020	No protocol usage, used zk-SNARKs but we only looked for zk-STARK in this extra search
Privacy-Preserving Post-quantum Credentials for Digital Payments	10.1007/978-3-031-32415-4_10	No protocol usage, only compares to zk-STARK
Queer In AI: A Case Study in Community-Led Participatory AI	10.1145/3593013.3594134	No protocol usage
Towards verifiable and privacy-preserving account model on a consortium blockchain based on zk-SNARKs	10.1007/s12083-023-01497-7	No protocol usage, used zk-SNARKs but we only looked for zk-STARK in this extra search